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Non-Adaptive Scheduling of Entanglement Purification for All-Photonic Quantum Repeaters

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Abstract

Long-distance quantum networks require repeaters that distribute high-fidelity Bell pairs despite photon loss and operational noise. In all-photonic quantum repeaters built from Half-Repeater Graph State (HRGS), entanglement purification can be applied at different stages, from resource generation to individual hops and end-to-end entanglement. The number of copies and circuit used may also vary between stages, making the choice of where and how to purify a combinatorial scheduling problem. The source papers demonstrate the potential of such scheduling through a single hand-picked example, but do not systematically explore the space of possible schedules.

This thesis represents each fixed non-adaptive purification schedule as a Directed Acyclic Graph (DAG) and evaluates it with a single bottom-up pass computing fidelity, success probability, latency, rate, and generation cost. The evaluator reproduces the source paper’s three canonical fidelity benchmarks to four significant figures. Three search tiers then construct and compare candidate schedules under a shared generation-cost budget $E_{\max}$: structured baseline enumeration, an exact span-partition dynamic program, and beam search with an additional same-span pumping operation. All candidates are structurally validated and evaluated using the same physical model throughout the protocol.

At the source papers’ reference configuration, a schedule found by search at the same generation cost as the hand-picked baseline improves the output rate by 2.5%. With a relaxed cost constraint, search finds a schedule requiring only 60% of the baseline generation cost while achieving a 52.8% higher output rate. More importantly, when inner-qubit error is introduced, the fixed baseline can fall below the required fidelity, whereas a schedule can be found by search at the same generation cost while remaining feasible. The same feasibility advantage is observed on average across independently randomized per-hop heterogeneous networks. Further experiments examine its dependence on noise level, timing, and chain length.

These results establish existence within the implemented schedule families and search limits and not global optimality: beam search and its pumping operation retain only a bounded frontier at each stage and are therefore only heuristic. Consequently, failing to find a feasible schedule does not rule out its existence, as it may lie outside the explored search frontier.

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Use of AI tools. AI tools were used during this work for identifying relevant research papers and suggesting directions for further reading, assisting with the formalization of the model, and supporting software development and debugging. They were also used as a writing aid for proofreading, improving clarity, and correcting non-native or unnatural phrasing. The research, implementation, analysis, and final content of this thesis remain the author's responsibility.

To my grandparents, whose lives and example have shaped the person I have become.

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Chapter 1

Introduction

1.1 Motivation

Realizing a Quantum Internet [1], [2], i.e. a network connecting quantum devices over long distances, would enable applications with no classical equivalent [3], [4]. Quantum Key Distribution (QKD) provides a shared secret key whose security is guaranteed unconditionally by the principles of quantum mechanics, independently of an eavesdropper’s computational capabilities [5]–[7]. Distributed Quantum Computing (DQC) lets several small quantum processors connected by such a network act together as one larger one [8], [9]. Finally, networked quantum sensing can improve measurement precision beyond what independent classically connected sensors can achieve [3], [10], [11]. These applications rely on distributed entanglement as a shared quantum resource; in the repeater architectures considered here, this resource is represented by high-fidelity Bell pairs, also called *ebits*. Direct photon transmission cannot supply this resource over long distances because channel loss grows exponentially with length [3]. Quantum repeaters address this limitation [12]; Chapter 2 reviews how successive repeater generations do so.

Regardless of repeater generation, the delivered entanglement is noisy. Entanglement purification is one standard way to turn such noisy entanglement into a usable resource: it consumes several independently generated copies of a shared pair and returns fewer copies of higher fidelity [13], [14]. Each round trades additional resource consumption for a probabilistic fidelity gain. In a multi-hop repeater chain, purification is not restricted to a single point in the protocol: its placement and number of rounds can be chosen as part of the overall protocol design. The number of copies consumed and the purification circuit may also vary from one stage to another. This thesis studies an all-photonic repeater architecture built from Repeater Graph States (RGSs) [15] (Chapter 3 introduces its physical building blocks), where purification may be applied during resource generation, after individual links are established, or after end-to-end entanglement is created. Choosing how to allocate a fixed purification budget across this chain is therefore a combinatorial scheduling problem. The central question is which schedule gives the best specified trade-off between rate, fidelity, and resource cost.

1.2 The Scheduling Gap

Two recent papers by Benchasattabuse, Hajdušek, and Van Meter establish the specific architecture and scheduling flexibility this thesis builds on. Throughout this thesis, [16] is called *the Bridging paper* and [17] is called *the Integrating paper*. Together, they are called *the source papers*.

The Bridging paper introduces the Half-Repeater Graph State (HRGS) building block: half of an RGS, generated and held at a single station rather than requiring the full state to exist in one place, allowing an end node to act as its own source of entanglement. This enables the all-photonic scheme to interoperate with conventional memory-equipped repeater nodes instead of requiring every intermediate station to be purely photonic [16]. The Integrating paper then shows that entanglement purification can be inserted into this architecture at any stage, generation, link, or end node, and

mixed freely between stages, provided it is executed *optimistically* [17], [18], i.e. without waiting for a classical heralding outcome between rounds. Chapter 3 formalises why this timing model is what makes purification worth inserting at all.

The Integrating paper demonstrates this flexibility with a single chosen schedule and identifies the search for a better one, under a fixed physical network and resource budget, as an open problem. It does not establish which allocation of a fixed budget performs best, or whether that answer changes with the noise model or chain length. This is the scheduling gap this thesis addresses. Given a fixed network configuration, the research problem is to select a schedule from the combinatorially many possibilities, which may differ in purification location, copy count, circuit, and order, and optimise it for a specified objective such as rate, fidelity, or cost. The working hypothesis is that systematic search finds schedules that improve on the hand-picked baseline at equal cost and, more importantly, retain feasibility in regimes where that baseline fails. This thesis tests that hypothesis across several network configurations beyond the source papers’ reference configuration. Chapters 4 and 5 formalise this schedule space and the algorithms that search it, respectively.

1.3 Research Objectives and Contributions

Addressing this gap first requires a precise object to search over, since neither source paper formalises the schedule space. Second, it requires search algorithms capable of exploring that exponentially large space. This thesis makes four concrete contributions toward that end.

1. A formal, Directed Acyclic Graph (DAG)-based model of the schedule space for HRGS-based quantum repeaters (Chapter 4) and a bottom-up evaluator that reproduces the source papers’ fidelity benchmarks for its raw, heralded, and optimistic end-node pumping schedules (defined in Chapter 5) to four significant figures.
2. A three-tier search framework: structured brute-force baselines, an exact span-partition dynamic program, and bounded same-span pumping and beam-search extensions (Chapter 5). Every candidate schedule is structurally validated and evaluated using the same physical model. At the Integrating paper’s reference configuration, the matched-cost result improves rate by 2.5%, while a budget-relaxed result reaches a rate of 6195.95 at cost 60, a 52.8% improvement while only using 60% of the paper baseline’s budget (Chapter 6).
3. An empirical study of when scheduling changes feasibility rather than only quality, varying outer- and inner-qubit noise, the number of Bell-State Measurement (BSM) arms per hop, hop count, memory decoherence, and generation timing (Chapters 6–7). Its strongest result is a budget-fixed feasibility advantage: under non-ideal inner-qubit error, the paper schedule falls below the $F \geq 0.9$ threshold while a searched schedule at the same budget still clears it.
4. A synthesis of when scheduling matters (Chapter 7): the advantage is largest when the budget is tight enough for purification choices to affect feasibility and rate, but not so restrictive that no schedule remains feasible. Within the searched families, the minimum feasible budget over $N = 10\text{--}28$ grows approximately as $E_{\max}^{\min} \approx 1.19 N^{1.67}$, suggesting a superlinear trend rather than the linear $10N$ budget assumed in the Integrating paper. This beam-search result is beam-width sensitive and should not be interpreted as a proven asymptotic bound.

Every result above only establishes existence within the schedule families actually searched, rather than providing a global optimality guarantee; Section 7.2 returns to this distinction.

Chapter 2

Related Work and State of the Art

This chapter reviews the three main research areas on which this thesis builds. Section 2.1 discusses quantum repeater protocols, Section 2.2 covers entanglement purification, and Section 2.3 presents RGS-based architectures. It then places the scheduling gap identified in Section 1.2 in the context of the existing state of the art, discussed in Sections 2.4 and 2.5.

2.1 Quantum Repeater Protocols

Sending a photonic qubit directly through optical fibre or free space is limited by the channel's transmissivity, i.e. the probability that a photon reaches the receiver. For a channel of length L and linear attenuation coefficient α , it is given by $\eta = e^{-\alpha L}$. At the typical telecom C-band attenuation of $\alpha_{\text{dB}} \approx 0.2$ dB/km [19], a 2000 km fibre link incurs 400 dB of loss, corresponding to a transmissivity of $\eta = 10^{-40}$. Thus, even with perfect sources and detectors, direct transmission cannot maintain practical rates over long distances [20]. More generally, the repeaterless secret-key capacity of a lossy channel, bounded by the Pirandola-Laurenza-Ottaviani-Banchi Bound (PLOB), has the same impractical exponential scaling in the high-loss regime [3], [21], [22].

A quantum repeater [12] divides a long channel into shorter elementary links, establishes entanglement over those links, and stitches them into an end-to-end pair through entanglement swapping [23], [24]. Repeater proposals are commonly grouped into three generations according to how they address photon loss and operational errors [3], [25]. First-generation (1G) repeaters use Heralded Entanglement Generation (HEG) and entanglement purification [13], [14] (Section 2.2), storing generated pairs in matter-qubit memories while other links are established [12], [26]. Second-generation (2G) repeaters replace purification with Quantum Error Correction (QEC) to correct operational errors, while still using heralding to handle photon loss [27], [28]. Third-generation (3G) repeaters use quantum error-correcting codes [29], [30] to protect against both loss and operational errors, avoiding heralding across every elementary link [15], [31].

The all-photonic quantum repeaters [15] studied in this thesis belong to the third generation but differ from proposals based on matter-qubit error correction. They encode the transmitted qubit in a multi-photon graph state [32]–[34]: the RGS [15]. Its tree encoding [35], [36] (Paragraph 2.2) provides loss tolerance while operating on flying qubits, removing the need for quantum memories at intermediate stations. Section 2.3 places this architecture among related proposals and shows its structure (Figure 2.1); Chapter 3 gives the mechanics required here.

A related but architecturally distinct proposal, the measurement-based quantum repeater [37], stores small pre-prepared entangled resource states [38]–[42] at each station and couples them to incoming photons using simple BSs, rather than constructing one graph state spanning the repeater chain [43]–[45]. Its emphasis on minimising the number of qubits held at each station provides a useful comparison for the resource costs studied in Chapter 6.

2.2 Entanglement Purification Protocols

Section 1.1 introduced the basic trade-off between the number of stored ebits and fidelity; this section covers the protocols that implement it.

The two foundational two-qubit protocols, both using only Local Operations and Classical Communication (LOCC), are BBPSSW [13], which operates on Werner states, and DEJMPS [14], which can directly purify general Bell-diagonal states, i.e. states where the four Bell states may have different probabilities, without first depolarizing the input state. Both consume two copies and return one, succeeding only for appropriate local measurement outcomes. Thus, purification improves the quality of an entangled pair by consuming additional entangled resources, establishing the resource-fidelity trade-off that is central to this thesis.

Entanglement pumping applies the same principle repeatedly: a working pair is purified against a sequence of freshly generated pairs, increasing its fidelity with each round [13], [14], [46]. Unlike a fixed purification circuit applied once, pumping accumulates purification over multiple rounds while retaining one working pair.

Later work treats the purification circuit itself as an optimisation problem rather than fixing it in advance. These approaches search over local Clifford operations to find shorter circuits with higher success probabilities and output fidelities than the BBPSSW/DEJMPS family [47]. This can be particularly beneficial when resources are finite and local operations are imperfect. Purification has also been extended from bipartite Bell pairs to multipartite graph states [48], [49]. The two-to-one purification circuits used by the Integrating paper (Section 3.3.1) combine these ideas for the RGS architecture rather than directly implementing BBPSSW or DEJMPS.

Photon loss tolerance Purification addresses noise, not photon loss. Tree encoding [35] provides a different form of redundancy: instead of consuming multiple copies to improve fidelity, it encodes one logical qubit in a tree-shaped graph state whose branching vectors determine how well it tolerates photon loss. Logical measurement outcomes can then be reconstructed from surviving branches. The approach was introduced for one-way measurement-based quantum computation [35], [38], [40] and later developed into a loss-tolerant quantum memory scheme [36], together with an analysis of its resource-efficiency threshold [50]. This tree encoding allows the inner qubits of the RGS to tolerate photon loss (Section 3.1.2, Figure 2.1).

2.3 Repeater Graph State Architectures

The RGS was proposed as the resource state for the first all-photonic quantum repeater [15]: a graph state whose outer qubits form entangling links to neighbouring stations while its inner qubits are tree-encoded [35], allowing the logical connection to survive photon loss [15]. Figure 2.1 illustrates this structure. Later work has focused on efficient RGS generation, with improved tolerance to photon loss and operational errors. The original proposal generates the graph state by fusing small entangled photon sets using linear optics [15], [22]. Deterministic emitter-based schemes instead build it directly from a small number of photon emitters, reducing the physical device count per station [51]–[53]. These approaches have also been demonstrated experimentally at small scale [54]. For loss and error tolerance, alternative graph topologies and adaptive measurement strategies can improve entanglement rates for a fixed loss and error budget [55]–[57]. Code-concatenated variants instead address both through two specialized codes rather than a single tree code [28].

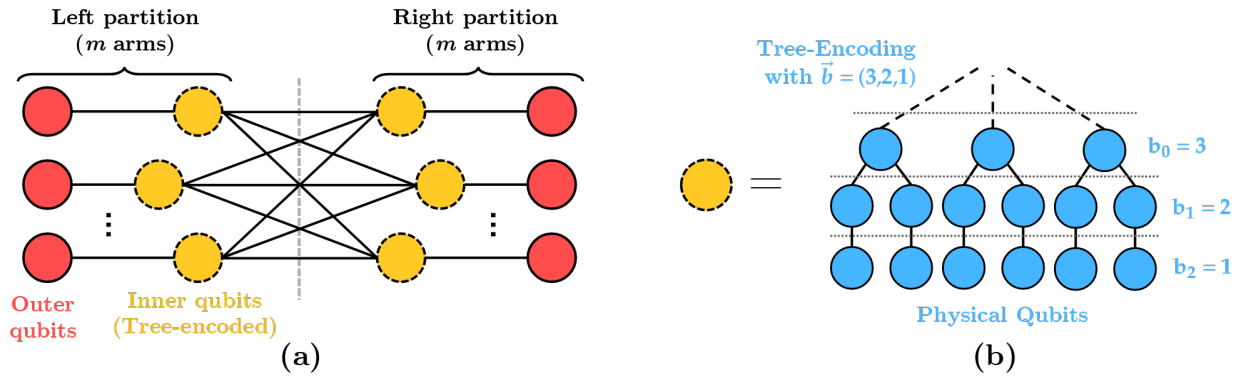


Figure 2.1: Structure of a Repeater Graph State (RGS) with m arms on each side. **(a)** The inner qubits (yellow) form two connected partitions, while each inner qubit is linked to an outer qubit (red), used to establish entanglement with a neighbouring station. The inner qubits are tree-encoded to improve the robustness of their logical measurements against photon loss [35]. **(b)** The corresponding tree encoding for branching vector $\vec{b} = (3, 2, 1)$, showing the physical qubits across successive tree levels.

The Bridging and Integrating papers, discussed in Section 1.2 and formalised in Chapter 3, provide the starting point for this thesis. The former introduces the Half-RGS (HRGS) building block [16], [58], [59], while the latter shows that purification can be incorporated using *optimistic* (blind) execution [17], [18]. This establishes that purification can be integrated, but not how it should be scheduled: where and how often to purify across a full chain remains an open problem, as discussed in Section 2.5.

2.4 Schedule Optimisation in Quantum Networks

Deciding when and where to apply entanglement purification is a separate problem, even when purification is available throughout a repeater chain. In adaptive schemes, a repeater node selects its next operation based on fidelity or heralding outcomes observed so far rather than following a fixed plan [60]. Reinforcement learning and other data-driven controllers have also been proposed for quantum-network control. One study, for example, reports substantial offline costs in richer settings, including exponentially growing state representations, training times of up to ~ 36 hours per configuration, and hyperparameter searches needed to outperform a heuristic baseline [61].

Optimistic purification has likewise been studied in generic quantum networks, independently of any RGS-specific construction [18], showing that deferred heralding can be useful beyond the architecture studied here. Another scheduling decision is how qubits of a logical channel are multiplexed across serial or parallel physical links, which affects end-to-end rate and fidelity [62]. These results demonstrate that the ordering and placement of operations can influence network performance, even when the underlying physical operations are already fixed.

2.5 Positioning of This Work

The work surveyed here generally optimises a fixed repeater component, such as a purification circuit or graph-state construction, or evaluates a specified scheduling policy. For the HRGS architecture studied in this thesis, however, these decisions are coupled: the choice of purification location affects the resources consumed, the resulting fidelity, and the timing of subsequent operations. No systematic search considers where to purify, how many copies to consume, which circuit to use, and in what order across a full chain. As Section 1.2 established, this is the gap addressed by this thesis.

Chapter 3

Background and Preliminaries

This chapter introduces the physical operations and mathematical quantities that Chapter 4 assembles into a searchable formal schedule model. Section 3.1 reviews Bell pairs and the HRGS repeater architecture proposed in the Bridging paper [16], whose fixed protocol-level operations form what Chapter 4 calls the *backbone layer*. Section 3.2 introduces the two quantities used to track a resource’s quality through a schedule: the error vector and the Z side-effect parity, together with the decoherence law that acts on the former while a resource is idle. Section 3.3 then covers the Integrating paper’s purification circuits and the distinction between heralded and optimistic execution, which determines their timing cost.

3.1 Quantum Networks and Repeaters

3.1.1 Bell Pairs and Fidelity

A Bell pair is an entangled two-qubit state shared between two nodes. The canonical Bell state is

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Its quality is measured by a single scalar, the fidelity $F = \langle \Phi^+ | \rho | \Phi^+ \rangle$, which quantifies how close the shared state ρ is to a perfect pair ($F_{\max} = 1$). Depolarising noise on the transmitted qubit as well as memory decoherence cause F to decrease, while photon loss instead reduces the probability of establishing a pair in the first place. Chapter 4 evaluates each finished schedule by its fidelity, rate, and resource cost for producing such a pair between the end nodes. Improving one of these quantities is meaningful only relative to this definition of F .

While Chapter 2 surveys why direct photonic transmission cannot distribute such pairs over long distances, we introduce here the elementary operation used to establish and extend links: a two-qubit Bell-State Measurement (BSM). A BSM performs a joint measurement on two incoming photons and identifies their Bell state through the measurement outcomes. On success, it effectively realizes entanglement swapping, extending the entanglement chain the network aims to create. Figure A.1 (Appendix A.1) illustrates this process. Using passive linear optics and no ancillary photons, a BSM succeeds with probability at most 50% per attempt, conditional on the photons arriving [16], [44]. This ceiling is why the architecture below builds redundancy into every hop.

3.1.2 Half-RGS Repeater Architecture

A *graph state* associates one qubit with every vertex of a graph $\mathcal{G} = (V, E)$, where V is the set of vertices and E is the set of edges connecting them: each qubit starts in state $|+\rangle$ and a controlled- Z gate is applied along every edge of the graph [32]–[34], [40]. The result is a multipartite entangled resource whose measurement outcomes at one vertex can be used, with appropriate local single-qubit operations, to establish entanglement between other vertices [37], [38], [40]. This is the underlying resource of the architectures surveyed in Section 2.3, including the one this thesis builds on.

A *repeater graph state* (RGS) is one such graph state, with $2m$ inner qubits and $2m$ outer qubits, where m is the number of outer-qubit arms on each side, as illustrated in Figure 2.1. It is proposed as the basis for a memory-free all-photonic quantum repeater [15]. The inner qubits form a complete bipartite (*biclique*) graph and are tree-encoded [35], allowing the logical measurements required by the protocol to be performed despite photon loss [50]. Each outer qubit is attached to one inner qubit and is consumed by a BSM with a neighbouring station. Such a network has two station types: Repeater Graph State Source (RGSS) nodes, which generate the RGS and hold it only until it is transmitted, and Advanced Bell-State Analyser (ABSA) nodes, which perform the BSMs between neighbouring stations and forward only the resulting entanglement [15], [16]. Figure 3.2 below shows how these two station types alternate along a full chain.

The Bridging paper introduces the *half repeater graph state* (HRGS), consisting of half of an RGS and held at a single station, avoiding the need to generate the full $2m$ -qubit state at once (Figure 3.1). Two HRGSs are combined into a full RGS through entanglement swapping, referred to as *Swap* hereafter and treated as a single operation throughout (Section 3.1.3). Because a HRGS only needs to be generated and briefly held, an end node can act as its own source: it generates a HRGS using its stationary anchor and emits the photonic qubits toward its neighbouring ABSA, rather than requiring a separate RGSS. This is what lets end-node memory scale additively with the number of hops instead of multiplicatively, the central resource-saving claim of the Bridging paper [16].

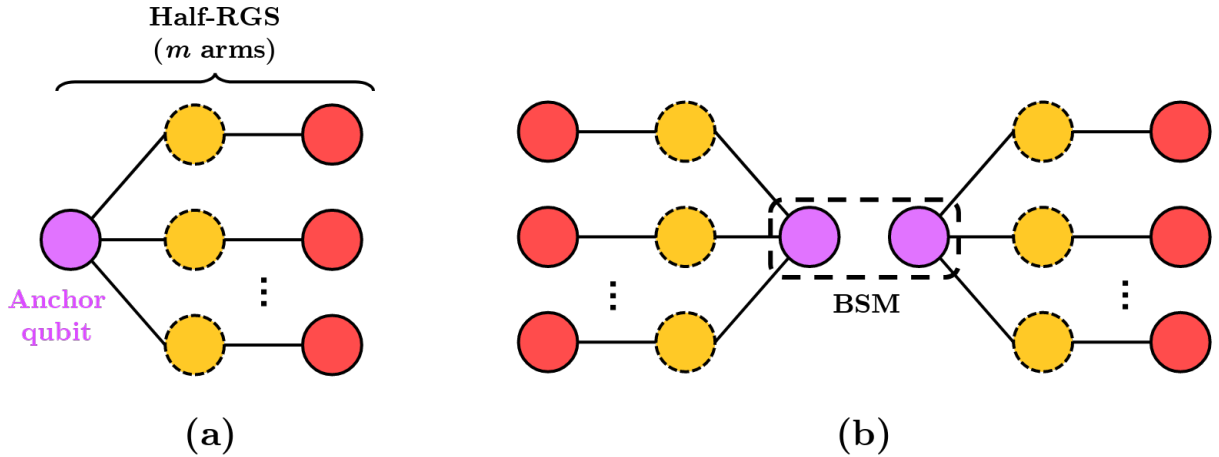


Figure 3.1: **(a)** An isolated HRGS, generated and held by a single station (an RGSS, or an end node acting as its own source). **(b)** Two such HRGSs, generated independently and possibly at different stations, combined into one full RGS by a controlled-phase (CZ) gate between their anchor qubits followed by a joint XX measurement on those same qubits. The measurement realizes entanglement swapping between the two HRGSs' remaining qubits, analogous to the effect of a BSM (Figure A.1) on two independent Bell pairs, but applied here to graph-state anchors.

An N -hop chain between two end nodes has $N + 1$ stations numbered $0, \dots, N$: stations 0 and N are the end nodes (Alice and Bob), and stations $1, \dots, N - 1$ are RGSS nodes. Every RGSS generates a full RGS and distributes its two halves to the ABSA on each side; an end node generates only the one HRGS it needs and sends its photonic qubits to its single neighbouring ABSA. Each hop i , the link between stations $i - 1$ and i , has one ABSA that receives outer qubits from both sides and performs the required BSMs, establishing entanglement one hop at a time until it spans the whole chain [16] (Figure 3.2).

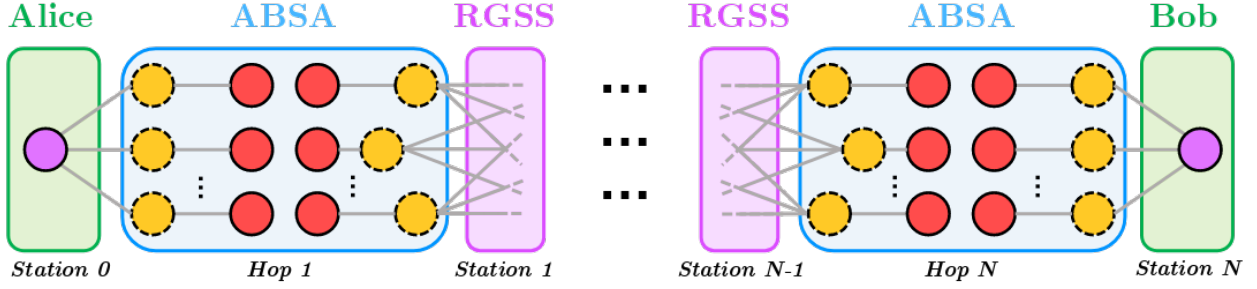


Figure 3.2: An N -hop network between Alice (station 0) and Bob (station N). Interior stations $1, \dots, N-1$ are RGSS nodes, each generating a full RGS and sending one half toward each neighbouring ABSA; the two end nodes instead generate only the single HRGS each needs (Section 3.1.2). Every hop’s ABSA receives outer qubits from both neighbouring stations and performs the corresponding BSMs; a successful BSM extends the established entanglement by one more hop, giving rise to the notion of *span* introduced next (Section 3.1.3).

3.1.3 Backbone Operations and Spans

Because the BSM’s success probability is capped at 50% (Section 3.1.1), each hop provisions redundant outer-qubit arms rather than a single pair. Chapter 4 uses k_i to denote the arm count at hop i . At every ABSA, once one arm’s BSM succeeds, that arm’s inner qubits are *kept*: the station *discards* the redundancy by measuring every other arm’s inner qubits in the Z basis [15], [16]. This *backbone selection* is forced by the protocol at every hop, on every trial. Figure A.2 in Appendix A.1 illustrates this process using a $N = 3$ -hop example.

A resource that has not yet crossed any hop – a freshly generated HRGS anchor or two such anchors combined by a Swap at the same station, for instance – is still *local* to its station of origin. Once a chain of BSMs and Swap operations connects stations across one or more hops, it forms a resource connecting stations a and b . Its *span* is written $\text{Span}(a, b)$, or simply (a, b) where no ambiguity arises. The symbol κ denotes the current stage of a resource, either local to a single station or spanning some interval of the chain; Chapter 4 gives this notation a full formal treatment. Two properties of spans matter for everything that follows:

1. A resource’s span is always a *contiguous interval of stations*, so that if it spans (a, b) then it also spans every intermediate station $a < i < b$. This is because every backbone operation that extends a span (a BSM or a Swap) connects resources across adjacent stations or at a shared endpoint.
2. A resource’s span is *independent of its history*. Two resources that have the same span (a, b) may have reached that state by entirely different routes, e.g. one purified at every hop and the other not at all, since all that matters is the current extent of the span.

Every BSM, arm measurement, and Swap operation can also produce a Pauli- Z side effect on the resulting resource, which must eventually be corrected, as described in Section 3.2.2. Resolving it requires only a small constant amount of classical communication per ABSA per trial, regardless of how deep the tree-encoded inner qubits are. This is because the side effects of each arm collapse to a small number of bits before the ABSA passes them on [16].

3.2 Error Representation

3.2.1 The Error Vector

Just like the two source papers [16], [17], this thesis models outer-photon depolarising noise, tree-encoded inner-qubit measurement error, and memory decoherence (Section 3.2.3) as Pauli channels acting on an ideal Bell pair. Conditional on a successful transmission, the resulting state remains diagonal in the Bell basis at every stage. It is therefore represented by a real normalized vector of four probabilities,

$$\mathbf{e} = (w, x, y, z)^\top, \quad w + x + y + z = 1, \quad (3.1)$$

corresponding to the Bell-diagonal density operator

$$\rho(\mathbf{e}) := w |\phi_{II}\rangle\langle\phi_{II}| + x |\phi_{ZI}\rangle\langle\phi_{ZI}| + y |\phi_{ZZ}\rangle\langle\phi_{ZZ}| + z |\phi_{IZ}\rangle\langle\phi_{IZ}|. \quad (3.2)$$

The four components are the probabilities of no error, a Z error on the first qubit only, a Z error on both qubits, and a Z error on the second qubit only, respectively [17]. The first component w is exactly the fidelity of Section 3.1.1 and is the quantity Chapter 4 reads off as the fidelity cost function $F(\Sigma; \mathcal{N})$. The same formalism describes a pre-transmission local pairing (an anchor and its own outer qubit, before either has crossed a hop) just as it does an established edge, which is what allows purification to be defined once and reused identically at every stage of the schedule (Section 3.3.2). This representation is closed under every operation the backbone and scheduling layers perform.

At the error-vector level, a BSM and a Swap operation apply the same bilinear composition rule to their two inputs:

$$\text{BSM}(\mathbf{e}_1, \mathbf{e}_2) = \begin{bmatrix} w_1 w_2 + x_1 x_2 + y_1 y_2 + z_1 z_2 \\ w_1 x_2 + x_1 w_2 + z_1 y_2 + y_1 z_2 \\ w_1 y_2 + y_1 w_2 + x_1 z_2 + z_1 x_2 \\ w_1 z_2 + z_1 w_2 + x_1 y_2 + y_1 x_2 \end{bmatrix}, \quad (3.3)$$

which again yields a Bell-diagonal vector [16], [17]. Purification (Section 3.3.2) and idle-time decoherence (Section 3.2.3) preserve the same property. Two independent contributions feed into $\mathbf{e}$ as a resource crosses a hop: an outer-qubit term from the depolarising rate e_d applied to the transmitted photon and an inner-qubit term from the tree-encoded measurement error rates $p_{\text{in}}^X, p_{\text{in}}^Z$, which contributes only to the independent x and z components, never to y [16]. Under otherwise-isotropic noise, this biases the accumulated error toward independent single-sided Z errors (x, z) over correlated two-sided ones (y), a bias that Section 3.3.1 shows the purification circuits can exploit.

3.2.2 Z Side-Effect Algebra

Independently of the error vector, every backbone operation (Section 3.1.3) may flip the sign of the surviving qubit's stabilizers, producing a Pauli- Z *side effect*. Unlike $\mathbf{e}$, a side effect is exactly known once its source measurement outcome is known. It therefore needs only a single bit $s \in \mathbb{F}_2$ per resource rather than a probability distribution. Side effects compose by eXclusive OR (XOR) whenever two resources are merged at a BSM, an arm measurement, or a Swap. This composition rule is exact, since a Pauli- Z commutes through every other operation in the model up to a sign that is tracked the same way [16]. The parity accumulated during HRGS generation, due to the sequence of emitter operations used to build its tree-encoded inner qubits, is incorporated into the same parity

bit before the HRGS is used. It therefore requires no separate bookkeeping. At the very end of a schedule, the total accumulated parity s^{total} at an end node is the XOR of every contributing Z side effect, and a single conditional correction $Z^{s^{\text{total}}}$ applied at that point is enough to recover the intended state.

This parity bookkeeping is separate from the error vector: s records the accumulated Pauli corrections associated with the measurement outcomes, while $\mathbf{e}$ describes the resource’s noise. For each arm, the individual measurement outcomes can be combined into their XOR parity, so only this single parity value needs to be communicated, regardless of the depth of the tree encoding [16]. The schedule evaluator tracks these parities to determine the correct Pauli frame, while candidate schedules are compared using their error vectors and the resulting cost functions.

3.2.3 Decoherence Model

While idle, a resource’s error vector relaxes toward the maximally mixed state at a rate set by the memory’s coherence time, $T_2 = 1/\gamma$, the standard coherence-time convention [63]:

$$\mathbf{e}(t) = \frac{1}{4}\mathbf{1} + e^{-\gamma t}(\mathbf{e}(t_0) - \frac{1}{4}\mathbf{1}).$$

This relaxation is applied locally and independently at every node of a schedule to the exact idle interval that node experiences. Consequently, two resources that wait for different durations are decohered before they are combined. Replacing those waits by one averaged duration after composition would generally produce a different error vector [16], [17]. In the implementation, these idle intervals are determined from the schedule’s operation times and the corresponding decoherence update is applied automatically during evaluation.

How much idle time a resource accumulates depends entirely on how a schedule is arranged, which is why the Integrating paper [17] singles out three canonical timing patterns that later serve, in Chapter 6, as a validation target for this thesis’ own cost functions:

- *Raw*: a chain with no purification at all, so every resource is used as soon as it is generated.
- *Baseline*: each purification round is heralded before the next round begins. This avoids speculative purification but incurs a separate classical confirmation delay after every round.
- *Optimistic*: generation and purification proceed back-to-back, with intermediate confirmation deferred until the very end of the sequence. Section 3.3.3 explains the resulting communication-time saving and its effect on rate.

3.3 Entanglement Purification

3.3.1 Purification Circuits: $\mathcal{C}_{YY}$, $\mathcal{C}_{ZX}$, and $\mathcal{C}_{XZ}$

Purification trades copies for quality, as surveyed in Section 2.2. In the RGS architecture considered here, a purification circuit acts on two resources at the same span κ (Section 3.1.3) through a probabilistic LOCC measurement. On success, it consumes the two inputs and leaves one output resource with an updated error vector; on failure, both inputs are discarded. The Integrating paper adapts this idea to the graph-state resources used here [17]. Three such circuits are available, distinguished by the stabilizer they jointly measure across the two input pairs: $\mathcal{C}_{ZX}$ detects a Z error on the second qubit of the pair, $\mathcal{C}_{XZ}$ detects a Z error on the first qubit [13], [14], and $\mathcal{C}_{YY}$ detects

the corresponding parity of Z errors across the four qubits involved [17]. Figure A.3 in Appendix A.1 shows the circuit-level implementation of these three purification circuits.

The measurement outcomes from the two parties are compared to determine whether the purification succeeds. Section 3.3.3 describes how this confirmation can be performed immediately or deferred. Because the inner-qubit noise term biases accumulated error toward independent single-sided errors (x, z) rather than correlated ones (y), purification sequences can exploit this bias by applying C_{YY} first, before switching to C_{ZX} or C_{XZ} [17] (Section 3.2.1).

3.3.2 Success Probabilities and Output States

The success probability and purified output vector for each circuit [17] are:

$$P_{C_{ZX}}(\mathbf{e}_1, \mathbf{e}_2) = (w_1 + x_1)(w_2 + x_2) + (z_1 + y_1)(z_2 + y_2), \quad (3.4)$$

$$P_{C_{XZ}}(\mathbf{e}_1, \mathbf{e}_2) = (w_1 + z_1)(w_2 + z_2) + (x_1 + y_1)(x_2 + y_2), \quad (3.5)$$

$$P_{C_{YY}}(\mathbf{e}_1, \mathbf{e}_2) = (w_1 + y_1)(w_2 + y_2) + (x_1 + z_1)(x_2 + z_2). \quad (3.6)$$

$$\text{Pur}_{C_{ZX}}(\mathbf{e}_1, \mathbf{e}_2) = \frac{1}{P_{C_{ZX}}} \begin{bmatrix} w_1 w_2 + z_1 z_2 \\ z_1 w_2 + w_1 z_2 \\ x_1 y_2 + y_1 x_2 \\ x_1 x_2 + y_1 y_2 \end{bmatrix}, \quad (3.7)$$

$$\text{Pur}_{C_{XZ}}(\mathbf{e}_1, \mathbf{e}_2) = \frac{1}{P_{C_{XZ}}} \begin{bmatrix} w_1 w_2 + x_1 x_2 \\ z_1 z_2 + y_1 y_2 \\ z_1 y_2 + y_1 z_2 \\ x_1 w_2 + w_1 x_2 \end{bmatrix}, \quad (3.8)$$

$$\text{Pur}_{C_{YY}}(\mathbf{e}_1, \mathbf{e}_2) = \frac{1}{P_{C_{YY}}} \begin{bmatrix} w_1 w_2 + y_1 y_2 \\ x_1 z_2 + z_1 x_2 \\ y_1 w_2 + w_1 y_2 \\ x_1 x_2 + z_1 z_2 \end{bmatrix}. \quad (3.9)$$

3.3.3 Heralded versus Optimistic Purification

In the *heralded* model, each purification round waits for a full classical round-trip ($2L/c$, where L is the end-to-end fiber length and c is the signal propagation speed) to confirm the previous round's success before the next round begins. The *optimistic* (or blind) model [17], [18] instead runs generation and purification back-to-back without intermediate confirmation: operations are executed speculatively, and their classical outcomes are compared only in a deferred communication phase at the end. This replaces the n_{pur} round-trip heraldings that heralded execution would otherwise require with a single final heralding phase, which is the main source of the rate advantage reported in [17].

Here, *optimistic* refers to when heralding is performed relative to purification in the schedule, rather than to a particular purification operation. The formal DAG representation in Section 4.3 makes this distinction explicit.

Chapter 4

The Formal Model

As established in Chapter 1, the Integrating paper [17] demonstrates schedules that combine generation, link-level, and end-node purification through a single hand-picked example, leaving the search for better schedules as an open problem. Section 4.1 introduces the network configuration used to define the physical setting of the problem, while Section 4.2 defines the resource state carried through a schedule. Section 4.3 then represents a schedule as a Directed Acyclic Graph (DAG) of operations, followed by the resource constraints in Section 4.4 and the optimization objectives in Section 4.5. Section 4.6 finally states the assumptions and limitations of the model.

The building blocks introduced in Chapter 3 – generation, swapping, and backbone selection – are reused without redefinition below. Together, they form the *backbone layer*: the fixed, protocol-determined process of establishing entanglement. The model introduced in this chapter represents the *scheduling layer* on top of it, specifying where purification is inserted, which circuit is used, the order in which operations are applied, and how resulting schedules are evaluated.

4.1 Network Configuration

A network configuration fixes the physical parameters and infrastructure that a schedule cannot change. As introduced in Section 3.1.2, stations are numbered $0, 1, \dots, N$ along the chain, so a hop is the link between consecutive stations $i - 1$ and i : an N -hop chain has N hops and $N + 1$ stations. With $i = 1, \dots, N$ indexing the hops, the configuration is the tuple

$$\mathcal{N} = \left(N, \{ \ell_i \}, \{ \vec{b}^{(i)} \}, \{ k_i \}, \{ (p_{\text{in},i}^X, p_{\text{in},i}^Z) \}, \{ \eta_i \}, e_d, \gamma, c, \tau_{\text{emit}} \right),$$

where, for each hop i , ℓ_i is its length; $\vec{b}^{(i)}$ is the list of tree-encoding branching vectors at successive levels; k_i is its number of redundant BSM arms; $p_{\text{in},i}^X$ and $p_{\text{in},i}^Z$ are its inner-qubit measurement error rates; and η_i is its photon survival probability. The remaining parameters are shared across the chain: e_d is the outer-photon depolarizing rate, γ is the memory dephasing rate, c is the signal propagation speed in the fiber, and τ_{emit} is an optional per-photon emission time used to model branching-dependent generation latency. Conditional on photon arrival, the per-arm BSM success probability is capped at 50% [16], [44] (Section 3.1.3). The factor η_i accounts separately for transmission loss, while k_i supplies redundancy against the measurement ceiling.

An *idealized network configuration* is one with identical (*uniform*) hop parameters ($\ell_i = \ell$, $\vec{b}^{(i)} = \vec{b}$, $k_i = k$ for all i) and noiseless inner-qubit measurements ($p_{\text{in},i}^X = p_{\text{in},i}^Z = 0$ for all i). Section 6.3.2 removes the assumption of noiseless inner qubits and refers back to this definition.

Unless stated otherwise, experiments start from the Integrating paper configuration, noted $\mathcal{N}_{\text{ref}}$. It has identically configured hops:

$$\begin{aligned} \forall i : \quad N = 10, \quad \ell_i = 2 \text{ km}, \quad \vec{b}^{(i)} = (16, 14, 1), \quad k_i = 18, \quad p_{\text{in},i}^X = p_{\text{in},i}^Z = 0 \\ e_d = 0.01, \quad \gamma = 0, \quad c = 2 \times 10^5 \text{ km/s}. \end{aligned}$$

The remaining per-hop parameter, η_i , is computed from ℓ_i using attenuation $\alpha = 0.2$ dB/km, giving $\eta_i = 10^{-\alpha \ell_i / 10} \approx 0.912$. The reference configuration leaves τ_{emit} unset, so the default evaluator adds no branching-dependent generation offset to the latency. Table B.1 in Appendix B.1 restates these reference values and lists the parameters varied in the experiments.

4.2 Representing a Partially-Built Bell Pair

Every resource produced while a schedule is evaluated, from a local HRGS to the final end-to-end pair, is represented by a state:

$$S = (\mathbf{e}, s, t, t_{\text{gen}}, \kappa, r, h).$$

The vector $\mathbf{e}$ is the error vector introduced in Section 3.2.1, and s is the Z side-effect parity from Section 3.2.2. The remaining fields describe the resource's state within the schedule:

- t and t_{gen} are the current simulated time and the time at which the resource was first generated, respectively. Their difference, $\Delta t_{\text{idle}} = t - t_{\text{gen}}$, records the elapsed waiting time relevant to the decoherence model of Section 3.2.3.
- $\kappa \in \mathcal{K}$ is the *span* of the resource, where $\mathcal{K} = \{\text{RGSS}\} \cup \{\text{Span}(a, b) : 0 \leq a < b \leq N\}$ is the set of possible stages for a given network configuration (defined formally in Section 4.3). It is either RGSS for a resource that has not yet crossed a hop, or $\text{Span}(a, b)$ for a resource connecting station a to station b .
- r counts how many purification rounds have already been applied to this branch.
- $h \in \{\text{pending}, \text{resolved}\}$ records whether the classical heralding signal for this resource has been received yet.

The field that most directly constrains the possible combinations in a schedule is κ : two states can only be combined by a purification circuit if they carry the *same* κ . Two branches with different histories, one purified twice and the other once for instance, can still be combined as long as both currently have the same span. This rule lets the model express schedules that mix purification strategies across different parts of the chain, including the type of schedule illustrated in the source paper's Fig. 4 [17].

4.3 Schedules as Directed Acyclic Graphs

A schedule is written as a pair $\Sigma = (T, \phi)$, where T is a rooted Directed Acyclic Graph (DAG) and ϕ is a labeling function

$$\phi : T \rightarrow \mathcal{K}, \quad \mathcal{K} = \{\text{RGSS}\} \cup \{\text{Span}(a, b) : 0 \leq a < b \leq N\},$$

assigning to every node of T the span κ of the state that node produces, using the same κ introduced in Section 4.2. Each node kind represents a distinct physical or logical action. The colours and shapes below form the legend used in the worked schedule of Figure 4.1:

- **GenNode** (light-blue ellipse): HRGS generation; these nodes are always leaves of T at $\kappa = \text{RGSS}$;
- **JoinNode** (orange box): the photonic BSM that joins the two RGSS-local HRGS resources on either side of one hop into a single-hop edge;
- **SwapNode** (green box): entanglement swapping, stitching two compatible spans (or two RGSS-local resources) into a wider one;
- **PurifyNode** (purple box): a two-to-one $\mathcal{C}_{YY}$, $\mathcal{C}_{ZX}$, or $\mathcal{C}_{XZ}$ circuit (Section 3.3.1);
- **HeraldNode** (yellow diamond): resolving the classical heralding signal for a branch;
- **IdleNode** (grey dashed box): an explicit wait, used only where the schedule's timing requires one;
- **PauliCorrectNode** (red double circle): the root of T at $\kappa = \text{Span}(0, N)$, representing the finished end-to-end pair after final Pauli correction.

Optimistic versus heralded schedules A schedule Σ is *fully optimistic* when no **HeraldNode** separates consecutive **PurifyNode** steps on any branch. In the optimistic schedules used here, all heralding is deferred to one **HeraldNode** directly below **PauliCorrectNode** at the root. A schedule is *fully heralded* when a **HeraldNode** is placed between consecutive purification rounds; each intermediate herald forces a classical round-trip ($2L/c$) before the next round can begin. Optimism is therefore a property of the schedule's timing structure, represented by the placement of **HeraldNodes**, rather than of an individual purification operation. The DAG representation can express mixed timing patterns, but the experiments compare fully optimistic and fully heralded schedules.

Schedule Legality Conditions A graph T is a *structurally legal* schedule when it has one **PauliCorrectNode** root, is acyclic, contains no unreachable nodes, and obeys the following stage constraints:

- (i) **GenNodes** may only appear at $\kappa = \text{RGSS}$;
- (ii) **JoinNodes** consume two RGSS-local resources associated with one hop and produce $\text{Span}(i, i+1)$;
- (iii) **PurifyNodes**' children must share the same κ , i.e. the span-matching rule stated informally in Section 4.2;
- (iv) **SwapNodes**' children must be either two RGS-local resources or two spans sharing an endpoint. For the latter, (a, b) and (b, c) with $a < b < c$, the spans need not correspond to the two neighbouring hops $(b-1, b)$ and $(b, b+1)$; they may each cover multiple hops, provided they meet at station b . The resulting resource has span (a, c) .
- (v) **IdleNode** and **HeraldNode** are single-child pass-through operations: neither an explicit wait nor resolving a heralding signal changes which stations a resource spans, so both copy their child's κ unchanged;
- (vi) the root must receive a state with $h = \text{resolved}$ (Section 4.2) at $\kappa = \text{Span}(0, N)$.

The implementation first validates the root, graph structure, and stage consistency, while the evaluator checks the corresponding physical conditions when processing each resource state.

The schedule representation was validated against the source paper’s own Fig. 4 schedule [17]. That demonstration schedule builds one Bell pair from purified individual links, a second from two purified 5-hop segments, and a third with no purification, before combining all three with a final purification round at the end nodes. It therefore combines branches purified to different degrees and built up in different ways. The present model can express this structure because κ may represent any span, while a `PurifyNode` requires only matching spans, not matching construction histories.

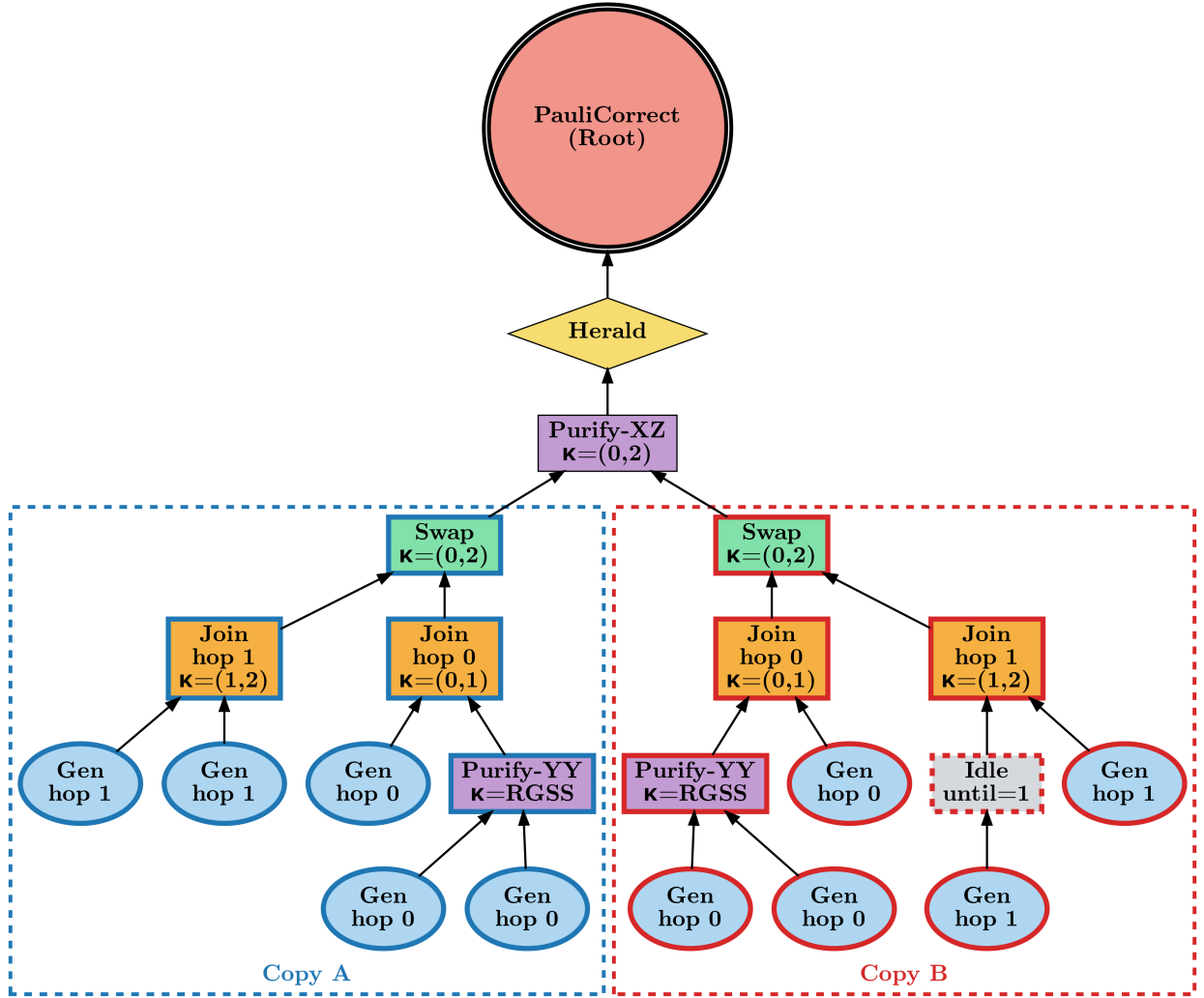


Figure 4.1: $N = 2$ example schedule Σ illustrating all seven node types ($C(\Sigma) = 10$ `GenNode` leaves). **Copy A** (left subtree): hop 0 applies `PurifyNode-YY` at $\kappa = \text{RGSS}$ before the outer-photon `JoinNode`; hop 1 is raw. **Copy B** (right subtree): identical except the right-side hop-1 `GenNode` is wrapped in an `IdleNode` (grey, dashed) modelling a synchronisation wait before the `JoinNode`. The two copies are first combined into $\kappa = (0, 2)$ by a `SwapNode`, then purified together by `PurifyNode-XZ`. The single `HeraldNode` placed after (not before) `PurifyNode-XZ` makes this an optimistic schedule (Section 4.3). `PauliCorrectNode` is the root at $\kappa = (0, 2)$ applying the final parity correction. Edges run bottom-to-top.

4.4 Resource Budget

A schedule is only meaningful together with a budget

$$B = (n_{\text{pur}}, E_{\text{max}}, M_{\text{max}}),$$

where n_{pur} specifies the number of extra HRGS copies generated for each purification round, E_{max} is the maximum number of **GenNode** leaves T is allowed to contain, and M_{max} is the maximum number of branches that may be under construction at the same time. This tuple B represents the overall resource budget for a given schedule.

Schedule Feasibility Conditions A schedule Σ is *formally feasible* with respect to B if it uses at most E_{max} **GenNode** leaves and never holds more than M_{max} branches open at once. The resource cost $C(\Sigma)$ used throughout Chapter 6 is simply the number of **GenNode** leaves in T , and every headline result enforces $C(\Sigma) \leq E_{\text{max}}$.

The implementation computes $M(\Sigma)$ as the minimum peak number of concurrently open branches over all legal execution orders, using the classical Sethi–Ullman recurrence for expression trees [64]. A *register* is interpreted here as an open branch: a resource that has been constructed but must remain available while another part of the schedule is evaluated. For each node v , let $r(v)$ denote the minimum number of registers required to evaluate the subtree rooted at v :

$$r(v) = \begin{cases} 1, & v \text{ is a } \mathbf{GenNode}, \\ r(c), & v \text{ is an } \mathbf{IdleNode} \text{ or } \mathbf{HeraldNode} \text{ with child } c, \\ \max\{r(c_1), r(c_2)\}, & r(c_1) \neq r(c_2), \\ r(c_1) + 1, & r(c_1) = r(c_2), \end{cases}$$

where the last two cases apply to binary nodes such as **JoinNode**, **SwapNode**, and **PurifyNode**. When the two child subtrees have different requirements, the smaller one can be evaluated first and its completed result held in a single register while the larger is evaluated, so the peak requirement is simply the larger of the two; when the requirements are equal, one subtree must be completed and held while the other is built, giving the additional +1. Thus $M(\Sigma) = r(\text{root}(T))$, computed in a single $O(|T|)$ bottom-up pass without any timing information. The pseudocode and complexity analysis are given in Appendix A.2. The schedule thus satisfies the concurrency budget when $M(\Sigma) \leq M_{\text{max}}$.

The source paper’s own budget convention is $E_{\text{max}} = 10N$: five HRGS copies per side, on both sides, for every hop. This convention is used as the reference budget throughout the experiments, and Section 6.5.1 asks directly whether it remains large enough as N grows.

4.5 Objective Functions and the Optimization Problem

Four metrics can be read from the finished schedule DAG by a single bottom-up pass:

- fidelity $F(\Sigma)$, the w component of the error vector at the root (Section 3.2.1);
- rate $R(\Sigma) = P_{\text{success}}/T_{\text{attempt}}$, where T_{attempt} is the deterministic wall-clock duration of one complete schedule attempt. Under the renewal model, a failed non-adaptive purification attempt restarts the entire schedule, so the expected wall-clock time per successful pair is $T_{\text{attempt}}/P_{\text{success}}$, whose reciprocal is the stated rate;

- resource cost $C(\Sigma)$, defined in Section 4.4;
- latency $L(\Sigma)$, the total duration of T .

The optimization problem this thesis addresses is therefore:

$$\Sigma^*(\mathcal{N}, B) = \underset{\substack{\Sigma \text{ legal w.r.t. } \mathcal{N} \\ \Sigma \text{ feasible w.r.t. } B}}{\arg \max} R(\Sigma; \mathcal{N}) \quad \text{s.t.} \quad F(\Sigma; \mathcal{N}) \geq F_{\min}.$$

In other words, among the structurally legal and formally feasible schedules, find the one with the highest rate while meeting a minimum fidelity requirement. This is the objective used in every headline result in Chapter 6 unless stated otherwise, with $F_{\min} = 0.9$ matching the source paper’s own threshold and $M_{\max} = \infty$ (i.e. no concurrency cap) unless a specific experiment states otherwise (Section 4.6).

The same four metrics support other research questions by changing which metric is maximized and which are constrained. Maximizing F subject to a rate floor asks for the best achievable quality at a given throughput, while minimizing C subject to floors on F and R asks for the cheapest schedule that still meets both requirements. Both variants were implemented as alternative objective presets. Section 6.4.1 shows that switching between these three presets requires no change to the search algorithms of Chapter 5: only the scoring function and feasibility constraints need to be changed.

4.6 Scope and Assumptions

This model makes three simplifying assumptions, which are described below.

First, only *non-adaptive* schedules are represented: T is fixed before execution starts and cannot branch on an intermediate measurement or heralding outcome. A fully adaptive scheme would require a different formulation, most naturally a Markov Decision Process (MDP) [65], [66]. This is left to future work (Section 8.2).

Second, $M_{\max}$ is computed exactly for any valid Σ (Section 4.4) and can be enforced as a feasibility constraint on the search objective. However, this constraint is only applied to complete candidate schedules when they are scored; it is not yet used for early pruning in the recursive span-partition search of Chapter 5, unlike $E_{\max}$. The reason is that $M(\Sigma)$ is not additive when independently constructed schedules for smaller spans are combined. Two schedules for disjoint spans may execute concurrently or sequentially, so the register count of their combined schedule depends on how the two spans are scheduled relative to each other. The register count of each smaller-span schedule alone is therefore insufficient to safely prune it. Every reported headline result consequently optimizes *without* an $M_{\max}$ cap.

Third, conditional on a successful transmission, the fidelity-noise model throughout this thesis is a Pauli channel [63], [67], [68], following the assumption made by the source papers. Photon loss instead reduces the success probability P_{success} rather than the fidelity F . Comparisons between schedules are therefore made under this specific fidelity-noise model. A different physical noise model could change the relative performance of the schedules, as discussed further in Section 7.4.

Chapter 5

Search Methods

Three search tiers share one schedule representation and one evaluator. Every returned candidate passes the structural checks of Section 4.3 and is evaluated by the bottom-up pass of Section 4.5; the tiers differ only in which legal DAGs they consider, never in how a returned DAG’s fidelity, success probability, latency, or cost is calculated.

Section 5.1 introduces the general concepts used throughout the search, followed by the shared evaluator in Section 5.2. The search is organised into four tiers with increasing scope but decreasing coverage of the schedule space: fixed baseline families in Section 5.3, the exact span-partition recurrence in Section 5.4, same-span pumping in Section 5.4, and beam search in Section 5.5. The latter is the heuristic used for the production-scale experiments in Chapter 6.

5.1 Preliminaries: Combinatorial Search Concepts

The schedule space defined in Section 4.3 is combinatorially large: even modest chains admit too many legal schedules to enumerate individually. The three tiers below rely on three standard tools from combinatorial optimisation, introduced briefly here before being applied to schedules.

Multi-objective optimisation and Pareto dominance A search problem is *multi-objective* when candidates are compared on several quantities at once, here cost C , fidelity F , and success probability P_{success} (Section 4.5). Candidate A *Pareto-dominates* B if A is at least as good on every quantity and strictly better on one; a *Pareto-optimal* candidate is dominated by none, and the *Pareto frontier* is the set of all such undominated candidates [69]–[71]. Retaining the whole frontier matters here because a locally inferior candidate can become the better choice once combined with another resource, as happens when two spans are swapped together.

Dynamic programming *Dynamic programming* decomposes a large problem into smaller overlapping subproblems that are solved once and reused wherever they recur (*memoized*) [72]. Section 5.4 applies this principle to schedules: candidate sub-schedules for a short span are computed once and reused when constructing longer spans.

Heuristic and beam search When candidates become too numerous to enumerate exhaustively, a *heuristic search* trades completeness for tractability by discarding some before they are fully explored. *Beam search* does this by keeping only a fixed number of candidates, the *beam width*, at each stage, ranked by a chosen rule [73]–[75]; a narrower beam is cheaper but more likely to discard a candidate that would have led to a better final schedule. Section 5.5 applies this to the span-partition recursion of Section 5.4.

5.2 The Schedule Evaluator

The evaluator makes one bottom-up pass over a schedule DAG. At each node it applies the corresponding physical operation to the current Pauli-channel state, synchronizes branches that become ready at different times, applies the resulting memory decoherence, and propagates side effects and timing. At the root it returns fidelity F , success probability P_{success} , resource cost C , and latency L . Rate is then $R = P_{\text{success}}/L$ under the full-restart model of Section 4.5.

Section 6.2 shows that this pass reproduces the source paper’s Fig. 5 fidelity numbers to four significant figures, confirming that the evaluator reproduces the source model correctly.

5.3 Fixed Schedule Families

The first tier enumerates four fixed structural families:

- a *single raw end-to-end chain*, with no purification;
- *end-node pumping*, in which n_{pur} independent raw chains are purified sequentially at the end nodes with a round-trip herald between rounds (*heralded* end-node pumping);
- its *optimistic counterpart*, deferring heralding until the end (*optimistic* end-node pumping);
- *uniform link-level* purification, using the same copy count and circuit at every hop.

For even N within budget, the Integrating paper’s hand-chosen schedule [17] is included as a fifth fixed comparison point.

Three of these families, the raw chain, heralded end-node pumping, and optimistic end-node pumping, are referred to throughout this thesis as the *canonical schedules*: they are the schedules whose fidelity is checked directly against the source paper’s published Fig. 5 benchmarks (Section 6.2).

Within each family, purification sequences use the three two-copy circuits $\mathcal{C}_{YY}$, $\mathcal{C}_{ZX}$, and $\mathcal{C}_{XZ}$ (Section 3.3.1). All 3^r circuit sequences are enumerated up to a configured round cutoff r_{max} ; above that cutoff, the implementation uses a small curated set. Thus, “brute force” denotes exhaustive enumeration within these predefined structural families and circuit grid, not enumeration of every legal schedule DAG.

These fixed families serve two purposes. First, they provide explicit baselines throughout Chapter 6: the raw chain and source-paper schedule serve as reference points, while uniform link-level purification provides a simple non-adaptive baseline without structural search. Second, their candidates are always retained alongside those produced by Second, their candidates are always kept alongside candidates from the broader searches, so they cannot be removed by pruning.

5.4 Span Dynamic Programming

For every span $\text{Span}(a, b)$, the second tier memoizes a Pareto frontier [69] of ways to create one resource over that span. A single-hop frontier ($b = a + 1$) contains the raw link and link-level purification choices. For a wider span, the recurrence tries every split point $a < m < b$ and combines each retained candidate from $\text{Span}(a, m)$ with each from $\text{Span}(m, b)$ using a Swap operation (Section 3.1.2). Each sub-span is computed once and reused in wider constructions. This avoids repeatedly constructing and evaluating complete end-to-end schedules, which is the main computational benefit of the dynamic program.

The frontier retained at each span is multi-objective over $(C, F, P_{\text{success}})$ (Pareto dominance, Section 5.1). Only non-dominated candidates are retained, following the standard Pareto-pruning rule [71]. This pruning is exact with respect to the three quantities tracked by the frontier but does not make the search exhaustive. The caps and heuristic retention rules introduced in the next two sections can still discard a non-dominated candidate before it reaches the end-to-end span.

Same-Span Entanglement Pumping The span-partition recurrence described above only builds a resource by swapping two *different* spans, so it cannot represent purifying a resource against a fresh, independently generated candidate at the *same* span κ . The second-tier extension allows this operation with limits on the number of copies and circuits used. The uncapped mode is used only for targeted validation, as it becomes impractical even at small N for large budgets (see Table B.6 in Appendix B.5).

Pumping candidates and ordinary Swap candidates share the same fixed-size frontier. As a result, adding pumping candidates can remove Swap candidates from the frontier before those candidates can be extended to a complete end-to-end schedule. Disabling pumping entirely reproduces pumping-free search behaviour; Section 6.3.1 reports both settings at the reference configuration since they can return different top-ranked schedules.

5.5 Beam Search

The third tier, beam search, reuses the same span recursion, node construction, and evaluator as Section 5.4, but caps every retained span frontier at a beam width w_{beam} (defaulted at $w_{\text{beam}} = 25$), bounding runtime and memory at the cost of possibly pruning a candidate a wider beam would keep. The cap is necessary in practice: an uncapped width-7 frontier already holds roughly 1,200 candidates, making an uncapped search through the paper’s $N = 10$ chain impractical. The same w_{beam} also caps the same-span pumping pool, so pumping and ordinary swap candidates share one set of slots at each span.

At each span, roughly half of the w_{beam} slots are reserved for the highest-fidelity candidates, while the rest favour candidates that meet F_{min} and have high success probability and low cost. This prevents the search from discarding link-level purification simply because it is less efficient locally. Such candidates may still be needed to maintain sufficient fidelity after several swaps.

Throughout this thesis, a *score* is the returned candidate’s value of whichever quantity the active objective maximizes (Section 4.5). Under the default rate-maximizing objective used for the beam-search benchmarks of Table B.6 in Appendix B.5, score is simply the candidate’s rate $R(\Sigma)$.

A different heuristic (e.g. simulated annealing or a genetic search) would need its own way of representing schedules and generating new candidates while respecting the legality conditions of Section 4.3. Beam search instead keeps the same span-partition representation and only changes how many candidates survive at each frontier, keeping every tier on one physical evaluation path and every run *deterministic*. As stated before, any given result thus only represents a valid candidate found by a bounded search, not a guarantee of global optimality.

Chapter 6

Experiments and Results

This chapter validates the evaluator against the Integrating paper’s [17] fidelity results, compares its schedule with schedules found by search under several network conditions, and evaluates cost-quality trade-offs, chain-length scaling, and timing sensitivity. Section 6.2 validates the physical evaluator against the source paper. Section 6.3 compares the paper schedule with schedules found by search, while Section 6.4 examines cost-quality trade-offs and alternative objectives. Section 6.5 studies how the required generation budget changes with chain length, and Section 6.6 evaluates memory dephasing and generation latency. Additional runtime, beam-width, and sweep results are reported in Appendix B. Unless the Bridging paper [16] is named explicitly, “the source paper” refers to the Integrating paper throughout this chapter. Additionally, *the optimizer* denotes the developed software [76] that implements the three search tiers of Chapter 5 and produces the reported search results.

6.1 Experimental Setup

Parameters and sweeps Unless stated otherwise, all experiments use the beam search with width $w_{\text{beam}} = 25$ (Section 5.5) together with the fixed schedule families of Section 5.3. They start from the reference configuration $\mathcal{N}_{\text{ref}}$ (Section 4.1), notably with $N = 10$, $e_d = 0.01$, $F_{\text{min}} = 0.9$, and $E_{\text{max}} = 100$. Each experiment changes only the parameters specified for that experiment; all remaining parameters are held fixed. A complete summary of the experimental parameters and sweep ranges is provided in Table B.1 (Appendix B.1).

Baselines Three fixed non-adaptive baselines, introduced in Section 5.3, are used throughout: the *raw chain*, with no purification; the *uniform link-level baseline*, which applies the same copy count and circuit sequence at every hop; and the *paper schedule*, the source paper’s hand-picked Fig. 4 example [17]. These are kept as separate baselines because the uniform link-level baseline applies the same purification strategy independently at every hop, whereas the paper schedule combines purification at different spans and stages of the network. The paper itself presents this schedule as a demonstration of scheduling flexibility rather than an optimized design.

Metrics As defined in Section 4.5, fidelity F is the Bell-pair fidelity after all purification, backbone, and correction operations, i.e. the w component of the error vector at the schedule root; rate is $R = P_{\text{success}}/T_{\text{attempt}}$, where P_{success} is the probability that a complete schedule attempt succeeds and T_{attempt} is its deterministic wall-clock duration; and resource cost C counts the number of HRGS copies – or equivalently **GenNode** leaves – consumed per trial. Two comparison modes recur throughout this chapter: *matched-cost*, in which the optimizer is constrained to $C = 100$, and *budget-relaxed*, in which it may choose any schedule satisfying $C \leq E_{\text{max}}$.

6.2 Physics Validation

The evaluator of Section 5.2 reproduces the source paper’s Fig. 5 [17] to four significant figures. At $e_d = 0.01$, the raw chain, heralded end-node pumping, and optimistic end-node pumping give

fidelities consistent with the corresponding values reported by the paper. This agreement validates the physical model used by the search tiers but does not establish search optimality. Appendix B.2 gives the corresponding comparison across the swept range of e_d .

6.3 Optimizer versus the Paper Schedule

This section first compares the search results with the source paper’s schedule at the reference configuration $\mathcal{N}_{\text{ref}}$, then examines the effect of non-ideal inner-qubit noise, outer-photon noise, and heterogeneous per-hop conditions.

6.3.1 Headline Comparison

At the reference configuration, the optimizer improves on the paper schedule both at the same cost and under the full resource budget. The matched-cost search achieves a higher rate, while the budget-relaxed search reaches a substantially higher rate at lower cost. The result obtained without enabling the same-span pumping operation is kept as a separate comparison. Detailed numerical results are reported in Table 6.1 below.

Schedule	C	F	Rate	Note
Paper schedule	100	0.93	4055.92	hand-picked
Matched-cost	100	0.92	4158.14	+2.5% rate
Budget-relaxed	60	0.91	6195.95	+52.8% rate, 60% of cost
Budget-relaxed, pumping disabled	50	0.90	6713.18	lower cost still

Table 6.1: Headline comparison at the reference configuration $\mathcal{N}_{\text{ref}}$ ($N = 10$, $e_d = 0.01$, $F_{\text{min}} = 0.9$, $E_{\text{max}} = 100$). The paper schedule is the hand-picked Fig. 4 construction from the Integrating paper [17]. Matched-cost search fixes $C = 100$; budget-relaxed search allows any schedule with $C \leq 100$. Rate is $R = P_{\text{success}}/T_{\text{attempt}}$.

6.3.2 Network Generalization

The optimizer was also evaluated under non-ideal inner-qubit errors and different numbers of redundant BSM arms. While the headline comparison uses the source paper’s idealized network configuration, the non-ideal configurations introduce physical constraints that can change which schedules remain feasible. At the same generation budget, the paper’s schedule falls below the fidelity threshold in both non-ideal configurations tested, whereas rerunning the optimizer under the corresponding physical parameters identifies *feasible schedules*. Figure 6.1 summarizes the resulting comparisons for the two BSM arm counts; the full numeric comparison is given in Table B.3 (Appendix B.3.1).

Rate values for an infeasible schedule are not directly comparable with those of a feasible schedule. Section 7.1 discusses the distinction between feasibility and rate more generally.

6.3.3 Per-Hop Heterogeneous Networks

So far, noise has only been varied uniformly across hops (Section 6.3.2). A real deployment, by contrast, will have different link lengths, hardware quality, and error rates from hop to hop. This section evaluates such heterogeneous networks by assigning each hop’s length, inner-qubit error rate, and arm count independently at random within the ranges used for this experiment. The search remains non-adaptive; only the supplied network configuration changes.

A five-hop network with one hop purposefully assigned a $12\times$ higher inner-qubit error rate than its neighbours illustrates the benefit of allocating purification resources *unevenly* across a

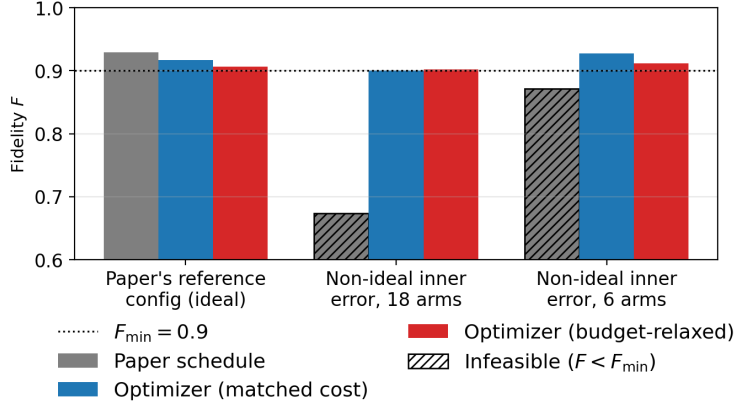


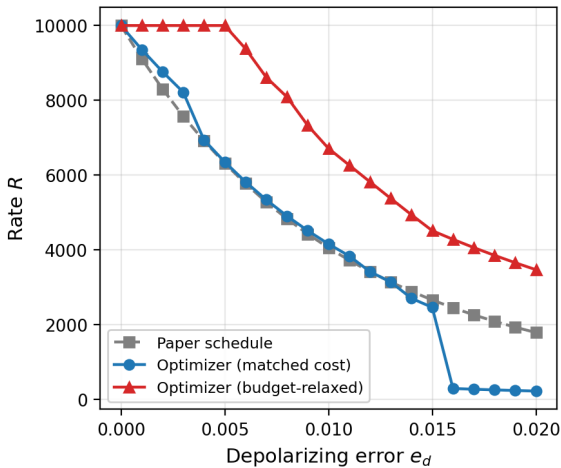
Figure 6.1: Fidelity by network configuration and schedule at $N = 10$, $e_d = 0.01$, $E_{\max} = 100$, $\gamma = 0$. Hatched bars fall below $F_{\min} = 0.9$. The paper schedule is infeasible under both non-ideal configurations; the schedules found by search remain feasible at the same budget.

heterogeneous chain, as detailed in Table B.4 (Appendix B.3.2).

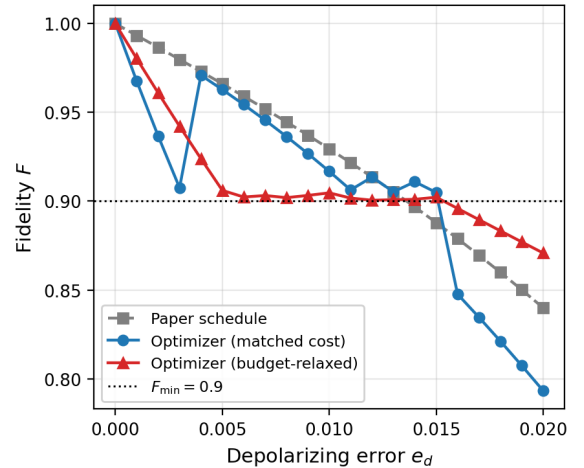
The same tendency holds over a sample of 100 independently generated random $N = 6$ networks (seeds 0–99, $e_d = 0.01$, $E_{\max} = 60$, $F_{\min} = 0.9$), reported in full in Appendix B.3.2. Among them, 80 were found to have at least one feasible uniform link-level candidate. The optimizer improves their rate by a mean of 20.37% and never performs worse in the tested sample (Section 7.1).

6.3.4 Noise Sensitivity

Sweeping e_d from 0 to 0.02 at $N = 10$ shows that the optimizer’s rate advantage over the uniform link-level baseline increases with noise. Figure 6.2 shows the rate (6.2a) and fidelity (6.2b) curves for the paper baseline, matched-cost optimizer, and budget-relaxed optimizer. At low noise the three variants remain close, with the rate curves separating notably around $e_d = 0.005$. The matched-cost curve represents the best cost-100 result at each point, not a single fixed schedule structure. Further interpretation is given in Section 7.1.



(a) Rate vs. e_d



(b) Fidelity vs. e_d

Figure 6.2: Rate and fidelity versus depolarizing error e_d at $N = 10$ and $E_{\max} = 100$ for the paper baseline, the optimizer at matched cost, and the budget-relaxed optimizer. The paper baseline’s higher fidelity in the right panel reflects unused surplus above the $F_{\min} = 0.9$ floor, purchased with cost and rate that the optimizer instead reallocates.

6.4 Cost-Quality Trade-offs

6.4.1 Pareto Frontiers

With a permissive objective ($F_{\min} = 0$, $E_{\max} = 150$), the beam search evaluates 180 candidates. Figure 6.3 shows two projections of the resulting Pareto frontier: fidelity vs. cost (6.3a) and fidelity vs. rate (6.3b). Dominated candidates and the non-dominated frontier under each pair of objectives are distinguished as described in the caption (Section 5.1, Pareto frontiers).

In the fidelity-cost projection, the paper schedule lies inside the frontier, which changes little once cost exceeds $C \approx 60$. In the fidelity-rate projection, the paper schedule is again inside the frontier, which trades off high fidelity against high rate. The paper schedule and the uniform link-level baseline are marked for reference.

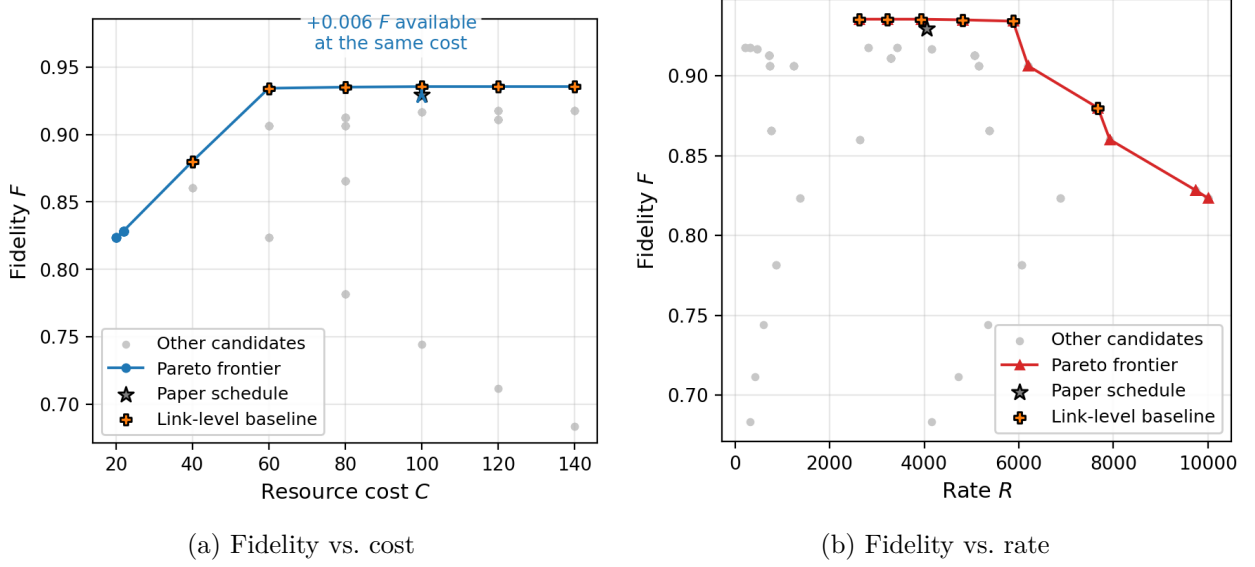


Figure 6.3: Pareto frontiers at $N = 10$, $e_d = 0.01$, $E_{\max} = 150$, over candidates evaluated under a permissive objective ($F_{\min} = 0$). Grey points are dominated; connected points form the non-dominated frontier, with the paper schedule and uniform link-level baseline marked for reference.

Alternative objectives Changing the objective from maximizing rate to maximizing fidelity or minimizing cost produces different top-ranked schedules without changing the search algorithm. Maximizing fidelity subject to a rate floor returns a higher-fidelity schedule; minimizing cost with only a fidelity floor can select a substantially lower-rate schedule. Adding an explicit rate floor changes that selection without modifying the search procedure; the complete numerical comparison is given in Appendix B.4.1.

6.5 Scaling with Chain Length

6.5.1 Budget Scaling

For N ranging from 10 to 28, the minimum $E_{\max}$ for which a schedule within the searched families reaches $F_{\min} = 0.9$ is found by dichotomy, rerunning the beam search at each trial budget. A descriptive power-law fit to the resulting points in log-log space gives

$$E_{\max}^{\min} \approx 1.192 N^{1.666},$$

with an exponent greater than one over the measured range. The paper’s budget is first exceeded at $N = 28$, while remaining sufficient at $N = 26$. Figure 6.4 shows the measured points together with the paper’s linear budget and the fitted power law. These fitted values describe the minimum budget found within the searched families at the chosen beam width, not a global lower bound or an asymptotic result (Section 4.6). Chapter 7 returns to this scope limitation (Section 7.2).

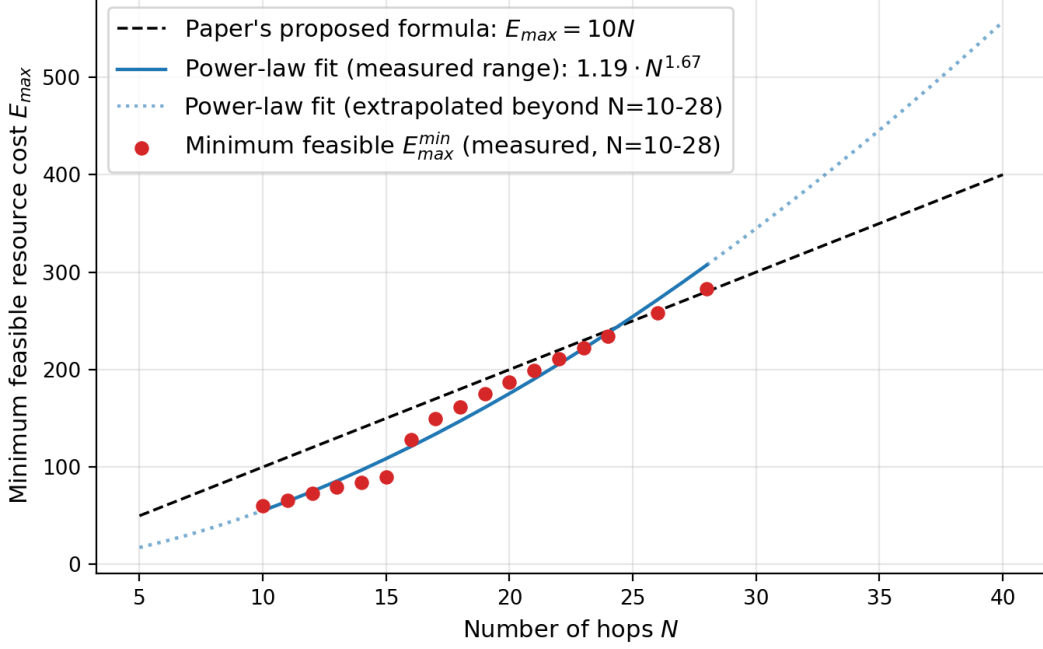


Figure 6.4: Minimum feasible $E_{\max}$ to reach $F_{\min} = 0.9$ vs. N at $e_d = 0.01$, compared with the paper’s linear $E_{\max} = 10N$ budget and the fitted power law $E_{\max}^{\min} \approx 1.192 N^{1.666}$. Solid markers and the solid fit cover the measured range ($N = 10-28$); the dotted extension is the same fit evaluated beyond that range, not additional data.

6.6 Timing Sensitivity

Memory dephasing and generation latency affect schedules through their timing structure, including the placement of `HeraldNodes` and `IdleNodes` (Section 4.3). Sweeping the memory dephasing rate γ leaves the raw chain and paper schedule unchanged over the tested range, while heralded end-node pumping degrades as γ increases because its sacrificial copies remain in memory during the round-trip heralding delays; the schedules selected by the optimizer remain flat over the tested range because the search favours optimistic schedules. Sweeping the generation latency τ_{emit} reduces the rate of every schedule. Detailed timing curves (Figure B.4) and the corresponding parameter sweeps are given in Appendix B.6.

Chapter 7

Discussion

Chapter 6 reported the results experiment by experiment. This chapter synthesizes them into five broader findings. Section 7.1 discusses when and why scheduling improves performance, Section 7.2 defines the scope of the resulting optimality claims, Section 7.3 discusses the heralded-versus-optimistic timing trade-off exploited by the searched schedules, and Section 7.4 summarizes the limitations of the model and search procedure.

7.1 When Scheduling Matters

The five findings below concern the *performance and feasibility advantages* obtained by searching over purification schedules. They generalize beyond the reference configuration $\mathcal{N}_{\text{ref}}$, subject to the search scope of Section 7.2.

Feasibility and rate advantages Under the idealized noise model tested by the source papers, the paper schedule and the searched schedules are all feasible, and the gap between them is a modest rate improvement (2.5% at matched cost, Section 6.3.1). Introducing inner-qubit error, which is not modelled in the source papers, can make the paper schedule fall below the fidelity floor. At the same generation cost, the optimizer’s search can instead select a feasible schedule (Section 6.3.2). In both cases, the mechanism is the same: the searched schedule *concentrates purification where noise has accumulated most*, which a uniform hand-designed schedule cannot do by construction.

The scheduling advantage generalizes beyond hand-picked configurations A sample of 100 random per-hop heterogeneous networks confirms the same advantage beyond a single hand-chosen configuration: the searched schedule beats a uniform convention in 80 of 100 cases, by a mean of 20.37% in rate, and never does worse (Section 6.3.3). One deliberately constructed network, with a single hop assigned a $12\times$ higher inner-qubit error rate than its neighbours, similarly causes the paper schedule to become infeasible while a feasible schedule can still be found by search, as detailed in Table B.4 (Appendix B.3.2).

The advantage concentrates in an intermediate regime The noise sweep (Section 6.3.4) and the Pareto frontier (Section 6.4.1) both show that scheduling matters most when the fidelity constraint becomes difficult to attain. On the other hand, scheduling barely matters when the fidelity floor is easy to satisfy, whether because noise is low or the allowed cost is generous. In that regime, most schedules are feasible. Their rates thus cluster together, and the searched frontier itself goes flat past a certain cost (already $C = 60$ at the reference noise level). The rate improvement over the uniform baseline increases from near zero at $e_d = 0$ to 18.2% at $e_d = 0.01$, with a clear difference emerging above about $e_d = 0.005$. However, this advantage does not persist at higher noise levels. With $e_d = 0.02$, both the uniform baseline and the budget-relaxed schedule fall below $F_{\text{min}} = 0.9$ once e_d reaches approximately 0.014–0.016. Beyond this point, neither schedule is feasible, so comparing their rates is no longer meaningful. Scheduling therefore matters most at intermediate noise levels, where the fidelity constraint is restrictive enough to distinguish between schedules but still allows feasible solutions.

In the noise sweep’s fidelity panel (Figure 6.2b), the paper baseline sits above both optimizer variants for nearly the whole range. This does **not** mean that it performs better than the optimizer: once $F_{\min} = 0.9$ is reached, additional fidelity provides **no benefit**. The paper schedule always spends the full $C = 100$ budget on a fixed circuit, while the budget-relaxed optimizer uses *less*, starting from $C = 20$ near $e_d = 0$ and increasing towards $C = 100$ as noise grows. It therefore accepts fidelity closer to the floor in exchange for higher success probability and rate, as shown in Figure 6.2a. At higher noise, this trade-off also extends the feasible range: the fixed paper schedule falls below $F_{\min} = 0.9$ by $e_d = 0.014$, while both optimizer variants remain feasible by re-optimizing the schedule as e_d increases.

The same constraint becomes harder to sustain with chain length The minimum generation cost required to achieve $F \geq 0.9$ grows faster than the linear $10N$ budget proposed in the source paper over the tested range, with a descriptive fit of $E_{\max}^{\min} \approx 1.19 N^{1.67}$ (Section 6.5.1). The fitted curve first exceeds the $10N$ budget at $N = 28$. This fit is descriptive rather than asymptotic, and the crossover is sensitive to the beam width, as discussed in Section 5.5. Even at $N = 10$, the paper’s proposed budget already exceeds the minimum required cost by a factor of 2.0–4.8 across the tested noise range. These observations reflect the same underlying effect seen in the noise sweep: *a fixed fidelity floor becomes harder to sustain as either the chain length or the noise increases*. Section 7.2 revisits the budget-scaling result below, where limitations of the searched families become important when interpreting this scaling.

The search accepts other objectives without requiring new algorithms None of the results above needed a different search procedure. Replacing the objective, rate, fidelity, or cost changes only which schedule is returned, not how the search finds it (Section 6.4.1), because candidate construction and evaluation are kept separate throughout Chapter 5.

7.2 Optimality Scope

The beam-search tier (Section 5.5) includes one same-span pumping operation, with the candidate frontier bounded by the beam width w_{beam} . All returned schedules are therefore valid constructions within the searched families, not proofs of global optimality. Finding the same best candidate when the search is run with two different beam widths provides evidence that the result is stable with respect to this parameter, but does not prove that no better candidate was pruned at either width. An uncapped beam width pumping mode (Section 5.4) is available but becomes impractical on larger instances, so it is used only for validation on smaller networks rather than as the default search method.

The *limited expressiveness* of the search space is demonstrated by the following example. The optimizer can only construct schedules from the primitive blocks and schedule families introduced in Chapter 5, and therefore cannot represent every structurally valid schedule. For example, it cannot construct a schedule by combining two independently constructed full-span candidates. At $N = 18$, such a manually constructed schedule fits within the Integrating paper’s budget of $E_{\max} = 10N = 180$, while the optimizer finds a schedule requiring an even smaller minimum budget (Section 6.5.1). The two results are therefore not in conflict: the latter concerns only schedules within the optimizer’s search space. More generally, the budget-scaling result gives the minimum budget among the searched families, not a lower bound on the budget required by all legal schedules.

The results of Chapter 6 should therefore distinguish between existence and non-existence claims. An existence claim, that a feasible schedule was found, holds for the searched model because the

schedule is *explicitly constructed*, validated, and evaluated by the shared physical model. A non-existence claim applies only to the families and search limits actually explored and does not prove that no legal schedule exists outside that space.

7.3 Heralded vs. Optimistic Execution

The timing results in Section 6.6 show that the relative benefit of optimistic and heralded execution depends on both *memory dephasing* and *generation latency*. The description of herald placement in Section 4.3 captures this timing trade-off. An optimistic sequence defers heralding until later in the sequence, eliminating intermediate classical round-trips but allowing subsequent operations to proceed before the outcome of an earlier purification is known. At the reference configuration, with no inner-qubit error and $\gamma = 0$, the optimizer *consistently favours optimistic candidates* because the reduced communication delay **outweighs** the loss in success probability. When $\gamma > 0$, heralded pumping is further penalized because its sacrificial copies remain in memory during each round-trip delay, increasing their exposure to decoherence.

Under the tested conditions, these effects converge in the same direction: reducing communication latency also reduces the time spent waiting in memory and therefore limits additional decoherence. The trade-off depends on the communication delay, memory coherence, and failure probabilities. With different physical parameters, intermediate heralding could instead be preferable. The observed preference should therefore be understood as a result of the configurations tested here, rather than as a general architectural rule.

7.4 Limitations

The results are subject to limitations in the schedule space, physical model, and bounded search.

1. *Non-adaptive schedules.* Only non-adaptive schedules are evaluated: the schedule is fixed before execution and cannot condition a later purification step on an intermediate outcome (Section 4.6). A fully adaptive strategy would require a different formulation, so the results bound only the non-adaptive schedule space.
2. *Memory footprint.* The search does not enforce limits on memory footprint. $M_{\max}$, the number of branches held in memory at once, is computed exactly for every candidate (Section 4.6), but only the generation-cost budget $E_{\max}$ constrains the search. A schedule feasible under $E_{\max}$ may therefore still require more simultaneously held branches than a deployment with a strict memory limit could support.
3. *Noise model.* The fidelity model assumes Pauli-channel noise, consistent with the source papers. Schedule comparisons are therefore internally consistent, but their relative performance could change under hardware dominated by different error processes. In particular, coupled loss, gate-error, and memory effects are not represented.
4. *Fixed beam width.* The search is sensitive to which construction types compete for retained slots. Enabling same-span pumping, for example, changes which candidate is returned at the reference configuration, so adding a construction can change previously reported results even at fixed beam width. Extensions should therefore include explicit before-and-after comparisons rather than assume that a new construction can only help.

Chapter 8

Conclusion

8.1 Summary

This thesis formalized the schedule space left informal by the source papers [16], [17] and searched it systematically, addressing the four contributions set out in Section 1.3.

The Directed Acyclic Graph (DAG)-based model of Chapter 4 provides a *common framework* for describing and evaluating purification schedules. Its evaluator reproduces the source paper’s fidelity results to four significant figures (Section 6.2), allowing different purification patterns, copy counts, and heralding placements to be evaluated under the same physical model. Built on this model, the three-tier search framework of Chapter 5 finds schedules with higher rates than the paper’s hand-picked example, both at matched cost and, more substantially, when the budget is relaxed (Section 6.3.1).

The strongest result is instead a *feasibility advantage*. Under inner-qubit error, which the source papers do not model, the papers’ fixed schedule can fall below $F \geq 0.9$ entirely, while a searched schedule at the same cost remains feasible (Section 6.3.2). The same pattern appears across 100 random per-hop heterogeneous networks and in a deliberately constructed noisy case (Section 6.3.3). These results show why schedule optimization becomes important when different parts of a chain experience different conditions: a fixed construction cannot redistribute purification effort, whereas a searched schedule can.

Chapter 7 combines this pattern with the noise and chain-length sweeps to identify a *common regime*: the advantage is largest when the fidelity constraint is strict enough to make the choice of schedule matter, but not so strict that all searched schedules become infeasible. This regime appears across all three tested axes: noise level, chain length, and network heterogeneity. The timing experiments provide a separate observation: the preferred heralding strategy depends on the balance between communication delay, memory dephasing, and purification success probability.

Results Existential Claims Limitations These results are existence claims within the searched schedule families, not proofs of global optimality over the full legal schedule space (Section 7.2). The finite beam width and restricted construction families *prevent concluding that no better schedule exists*. The excluded-construction check discussed in Section 7.2 illustrates this directly: it identifies a legal feasible schedule that no search tier could ever come up with because its construction lies outside their search spaces, regardless of how good the schedules found within those spaces are.

Taken together, the results support the working hypothesis stated in Section 1.2: *systematic search can improve on the paper’s hand-picked baseline at equal cost and, most notably, retain feasibility where that baseline fails*. The practical implication is that treating purification scheduling as an optimization problem can reveal useful constructions that a fixed uniform design would miss.

8.2 Future Work

The results also point to several extensions that would broaden the scope of the optimizer and reduce the gap between the current model and a deployable network.

- *Adaptive scheduling.* Conditioning later purification decisions on earlier measurement or heralding outcomes, within a single execution, could adapt to the realized state of the network rather than committing to a fixed sequence in advance. As noted in Section 4.6, this would most naturally be formulated as an MDP [65], [66], extending the current non-adaptive schedule space and potentially avoiding purification steps that become unnecessary after an intermediate outcome. Reinforcement learning is one candidate solution method, though the state and action spaces would need to be redefined carefully for this setting.
- *Online optimization.* Rather than adapting within a single execution, the search could be re-run periodically as noise and link quality change over time. In this thesis, the schedule is chosen once and then kept fixed. Re-running the search would allow the schedule to adapt to changing physical conditions, particularly in heterogeneous networks where different hops may degrade at different rates.
- *Multi-path routing.* Extending the DAG formalism from a single repeater chain to tree or mesh networks would require a more general connectivity representation and a corresponding extension of the current span-based search. In such networks, purification scheduling and route selection would become coupled decisions: the optimizer could potentially choose not only where and when to purify, but also which physical path or set of paths should be used to establish the end-to-end pair.
- *Beyond the Pauli-channel model.* Hardware-specific noise models, including more detailed photon-loss and gate-error processes, would let the optimizer be evaluated against a broader range of physical implementations and could change the relative ranking of candidate schedules.

Overall, this work provides a starting point for systematic purification-schedule optimization. Even within a restricted non-adaptive space, purification placement and allocation affect both rate and the ability to meet a target fidelity. Extending the approach to adaptive decisions, time-varying networks, richer topologies, and more realistic noise models would test whether these gains persist under more realistic conditions.

Code and data availability The evaluator, the three search tiers of Chapter 5, and every experiment script behind the tables and figures in this chapter and its appendix are openly available at [76].

Bibliography

- [1] R. Van Meter, *Quantum Networking* (Networks and Telecommunications Series). Hoboken, NJ: Wiley, 2014, 1 p. DOI: [10.1002/9781118648919](https://doi.org/10.1002/9781118648919).
- [2] W. Kozłowski, S. Wehner, R. V. Meter, *et al.*, “Architectural Principles for a Quantum Internet,” Internet Engineering Task Force, Request for Comments RFC 9340, Mar. 2023, 37 pp. DOI: [10.17487/RFC9340](https://doi.org/10.17487/RFC9340).
- [3] K. Azuma, S. E. Economou, D. Elkouss, *et al.*, “Quantum repeaters: From quantum networks to the quantum internet,” *Reviews of Modern Physics*, vol. 95, no. 4, p. 045006, Dec. 20, 2023. DOI: [10.1103/RevModPhys.95.045006](https://doi.org/10.1103/RevModPhys.95.045006). arXiv: [2212.10820](https://arxiv.org/abs/2212.10820) [quant-ph].
- [4] M. Hajdušek and R. V. Meter. “Quantum Communications.” arXiv: [2311.02367](https://arxiv.org/abs/2311.02367) [quant-ph]. (Nov. 4, 2023), [Online]. Available: <http://arxiv.org/abs/2311.02367>, pre-published.
- [5] C. H. Bennett, “Quantum cryptography using any two nonorthogonal states,” *Physical Review Letters*, vol. 68, no. 21, pp. 3121–3124, May 25, 1992. DOI: [10.1103/PhysRevLett.68.3121](https://doi.org/10.1103/PhysRevLett.68.3121).
- [6] C. H. Bennett and G. Brassard, “Quantum cryptography: Public key distribution and coin tossing,” *Theoretical Computer Science*, vol. 560, pp. 7–11, Dec. 2014. DOI: [10.1016/j.tcs.2014.05.025](https://doi.org/10.1016/j.tcs.2014.05.025). arXiv: [2003.06557](https://arxiv.org/abs/2003.06557) [quant-ph].
- [7] V. Scarani, H. Bechmann-Pasquinucci, N. J. Cerf, M. Dušek, N. Lütkenhaus, and M. Peev, “The security of practical quantum key distribution,” *Reviews of Modern Physics*, vol. 81, no. 3, pp. 1301–1350, Sep. 29, 2009. DOI: [10.1103/RevModPhys.81.1301](https://doi.org/10.1103/RevModPhys.81.1301).
- [8] J. I. Cirac, A. K. Ekert, S. F. Huelga, and C. Macchiavello, “Distributed quantum computation over noisy channels,” *Physical Review A*, vol. 59, no. 6, pp. 4249–4254, Jun. 1, 1999. DOI: [10.1103/PhysRevA.59.4249](https://doi.org/10.1103/PhysRevA.59.4249).
- [9] J. C. Boschero, N. M. P. Neumann, W. van der Schoot, T. Sijpesteijn, and R. Wezeman, “Distributed Quantum Computing: Applications and Challenges,” in vol. 1423, 2025, pp. 100–116. DOI: [10.1007/978-3-031-92602-0_6](https://doi.org/10.1007/978-3-031-92602-0_6). arXiv: [2410.00609](https://arxiv.org/abs/2410.00609) [quant-ph].
- [10] T. J. Proctor, P. A. Knott, and J. A. Dunningham. “Networked quantum sensing.” arXiv: [1702.04271](https://arxiv.org/abs/1702.04271) [quant-ph]. (Feb. 14, 2017), [Online]. Available: <http://arxiv.org/abs/1702.04271>, pre-published.
- [11] T. J. Proctor, P. A. Knott, and J. A. Dunningham, “Multiparameter Estimation in Networked Quantum Sensors,” *Physical Review Letters*, vol. 120, no. 8, p. 080501, Feb. 21, 2018. DOI: [10.1103/PhysRevLett.120.080501](https://doi.org/10.1103/PhysRevLett.120.080501).
- [12] H.-J. Briegel, W. Dür, J. I. Cirac, and P. Zoller. “Quantum repeaters for communication.” arXiv: [quant-ph/9803056](https://arxiv.org/abs/quant-ph/9803056). (Mar. 20, 1998), [Online]. Available: <http://arxiv.org/abs/quant-ph/9803056>, pre-published.
- [13] C. H. Bennett, G. Brassard, S. Popescu, B. Schumacher, J. A. Smolin, and W. K. Wootters, “Purification of Noisy Entanglement and Faithful Teleportation via Noisy Channels,” *Physical Review Letters*, vol. 76, no. 5, pp. 722–725, Jan. 29, 1996. DOI: [10.1103/PhysRevLett.76.722](https://doi.org/10.1103/PhysRevLett.76.722).
- [14] D. Deutsch, A. Ekert, R. Jozsa, C. Macchiavello, S. Popescu, and A. Sanpera, “Quantum Privacy Amplification and the Security of Quantum Cryptography over Noisy Channels,” *Physical Review Letters*, vol. 77, no. 13, pp. 2818–2821, Sep. 23, 1996. DOI: [10.1103/PhysRevLett.77.2818](https://doi.org/10.1103/PhysRevLett.77.2818).
- [15] K. Azuma, K. Tamaki, and H.-K. Lo, “All photonic quantum repeaters,” *Nature Communications*, vol. 6, no. 1, p. 6787, Apr. 15, 2015. DOI: [10.1038/ncomms7787](https://doi.org/10.1038/ncomms7787). arXiv: [1309.7207](https://arxiv.org/abs/1309.7207) [quant-ph].
- [16] N. Benchasattabuse, M. Hajdušek, and R. Van Meter, “Bridging All-Photonic and Memory-Based Quantum Repeaters,” *IEEE Transactions on Quantum Engineering*, vol. 7, pp. 1–15, 2026. DOI: [10.1109/TQE.2026.3653126](https://doi.org/10.1109/TQE.2026.3653126).
- [17] N. Benchasattabuse, M. Hajdušek, and R. V. Meter, “Integrating Entanglement Purification into All-Photonic Quantum Repeaters,” in *2025 IEEE International Conference on Quantum Computing and Engineering (QCE)*, Aug. 30, 2025, pp. 885–895. DOI: [10.1109/QCE65121.2025.00100](https://doi.org/10.1109/QCE65121.2025.00100). arXiv: [2504.18121](https://arxiv.org/abs/2504.18121) [quant-ph].

- [18] M. Mobayenjarihani, G. Vardoyan, and D. Towsley, "Optimistic Entanglement Purification in Quantum Networks," in *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)*, Bellevue, WA, USA: IEEE, Sep. 17, 2023, pp. 1143–1153. DOI: [10.1109/QCE57702.2023.00129](https://doi.org/10.1109/QCE57702.2023.00129).
- [19] "Recommendation ITU-T G.652 (08/2024) - Characteristics of a single-mode optical fibre and cable," International Telecommunication Union (ITU). (Nov. 22, 2024), [Online]. Available: <https://www.itu.int/epublications/publication/itu-t-g-652-2024-08-characteristics-of-a-single-mode-optical-fibre-and-cable>.
- [20] N. Sangouard, C. Simon, H. de Riedmatten, and N. Gisin, "Quantum repeaters based on atomic ensembles and linear optics," *Reviews of Modern Physics*, vol. 83, no. 1, pp. 33–80, Mar. 21, 2011. DOI: [10.1103/RevModPhys.83.33](https://doi.org/10.1103/RevModPhys.83.33).
- [21] S. Pirandola, R. Laurenza, C. Ottaviani, and L. Banchi, "Fundamental limits of repeaterless quantum communications," *Nature Communications*, vol. 8, no. 1, p. 15043, Apr. 26, 2017. DOI: [10.1038/ncomms15043](https://doi.org/10.1038/ncomms15043).
- [22] M. Pant, H. Krovi, D. Englund, and S. Guha, "Rate-distance tradeoff and resource costs for all-optical quantum repeaters," *Physical Review A*, vol. 95, no. 1, p. 012304, Jan. 4, 2017. DOI: [10.1103/PhysRevA.95.012304](https://doi.org/10.1103/PhysRevA.95.012304). arXiv: [1603.01353](https://arxiv.org/abs/1603.01353) [quant-ph].
- [23] M. Żukowski, A. Zeilinger, M. A. Horne, and A. K. Ekert, "'Event-ready-detectors' Bell experiment via entanglement swapping," *Physical Review Letters*, vol. 71, no. 26, pp. 4287–4290, Dec. 27, 1993. DOI: [10.1103/PhysRevLett.71.4287](https://doi.org/10.1103/PhysRevLett.71.4287).
- [24] A. Grudka, M. Horodecki, P. Horodecki, R. Horodecki, and M. Piani, "Entanglement-swapping boxes and their communication properties," *Physical Review A*, vol. 77, no. 6, p. 060307, Jun. 18, 2008. DOI: [10.1103/PhysRevA.77.060307](https://doi.org/10.1103/PhysRevA.77.060307). arXiv: [quant-ph/0703095](https://arxiv.org/abs/quant-ph/0703095).
- [25] S. Muralidharan, L. Li, J. Kim, N. Lütkenhaus, M. D. Lukin, and L. Jiang, "Efficient long distance quantum communication," *Scientific Reports*, vol. 6, no. 1, p. 20463, Feb. 15, 2016. DOI: [10.1038/srep20463](https://doi.org/10.1038/srep20463). arXiv: [1509.08435](https://arxiv.org/abs/1509.08435) [quant-ph].
- [26] L.-M. Duan, M. Lukin, I. Cirac, and P. Zoller, "Long-distance quantum communication with atomic ensembles and linear optics," *Nature*, vol. 414, no. 6862, pp. 413–418, Nov. 2001. DOI: [10.1038/35106500](https://doi.org/10.1038/35106500). arXiv: [quant-ph/0105105](https://arxiv.org/abs/quant-ph/0105105).
- [27] L. Jiang, J. M. Taylor, K. Nemoto, W. J. Munro, R. V. Meter, and M. D. Lukin, "Quantum Repeater with Encoding," *Physical Review A*, vol. 79, no. 3, p. 032325, Mar. 20, 2009. DOI: [10.1103/PhysRevA.79.032325](https://doi.org/10.1103/PhysRevA.79.032325). arXiv: [0809.3629](https://arxiv.org/abs/0809.3629) [quant-ph].
- [28] K. J. Wo, G. Avis, F. Rozpędek, *et al.*, "Resource-efficient fault-tolerant one-way quantum repeater with code concatenation," *npj Quantum Information*, vol. 9, no. 1, p. 123, Dec. 12, 2023. DOI: [10.1038/s41534-023-00792-8](https://doi.org/10.1038/s41534-023-00792-8).
- [29] E. Knill and R. Laflamme, "A Theory of Quantum Error-Correcting Codes," *Physical Review Letters*, vol. 84, no. 11, pp. 2525–2528, Mar. 13, 2000. DOI: [10.1103/PhysRevLett.84.2525](https://doi.org/10.1103/PhysRevLett.84.2525). arXiv: [quant-ph/9604034](https://arxiv.org/abs/quant-ph/9604034).
- [30] D. Gottesman. "Stabilizer Codes and Quantum Error Correction." arXiv: [quant-ph/9705052](https://arxiv.org/abs/quant-ph/9705052). (May 28, 1997), [Online]. Available: <http://arxiv.org/abs/quant-ph/9705052>, pre-published.
- [31] A. G. Fowler, D. S. Wang, C. D. Hill, T. D. Ladd, R. V. Meter, and L. C. L. Hollenberg, "Surface code quantum communication," *Physical Review Letters*, vol. 104, no. 18, p. 180503, May 6, 2010. DOI: [10.1103/PhysRevLett.104.180503](https://doi.org/10.1103/PhysRevLett.104.180503). arXiv: [0910.4074](https://arxiv.org/abs/0910.4074) [quant-ph].
- [32] H. J. Briegel and R. Raussendorf, "Persistent Entanglement in Arrays of Interacting Particles," *Physical Review Letters*, vol. 86, no. 5, pp. 910–913, Jan. 29, 2001. DOI: [10.1103/PhysRevLett.86.910](https://doi.org/10.1103/PhysRevLett.86.910).
- [33] M. Hein, W. Dür, J. Eisert, R. Raussendorf, M. Van den Nest, and H. J. Briegel. "Entanglement in Graph States and its Applications." version 1. (2006), [Online]. Available: <https://arxiv.org/abs/quant-ph/0602096>, pre-published.
- [34] M. Hein, J. Eisert, and H. J. Briegel, "Multi-party entanglement in graph states," *Physical Review A*, vol. 69, no. 6, p. 062311, Jun. 9, 2004. DOI: [10.1103/PhysRevA.69.062311](https://doi.org/10.1103/PhysRevA.69.062311). arXiv: [quant-ph/0307130](https://arxiv.org/abs/quant-ph/0307130).
- [35] M. Varnava, D. E. Browne, and T. Rudolph, "Loss Tolerance in One-Way Quantum Computation via Counterfactual Error Correction," *Physical Review Letters*, vol. 97, no. 12, p. 120501, Sep. 20, 2006. DOI: [10.1103/PhysRevLett.97.120501](https://doi.org/10.1103/PhysRevLett.97.120501).
- [36] M. Varnava, D. Browne, and T. Rudolph, "Loss Tolerant Linear Optical Quantum Memory By Measurement Based Quantum Computing," *New Journal of Physics*, vol. 9, no. 6, pp. 203–203, Jun. 29, 2007. DOI: [10.1088/1367-2630/9/6/203](https://doi.org/10.1088/1367-2630/9/6/203). arXiv: [0705.1904](https://arxiv.org/abs/0705.1904) [quant-ph].

- [37] M. Zwerger, W. Dür, and H. J. Briegel, “Measurement-based quantum repeaters,” *Physical Review A*, vol. 85, no. 6, p. 062326, Jun. 27, 2012. DOI: [10.1103/PhysRevA.85.062326](https://doi.org/10.1103/PhysRevA.85.062326). arXiv: [1204.2178](https://arxiv.org/abs/1204.2178) [quant-ph].
- [38] R. Raussendorf, D. E. Browne, and H. J. Briegel, “The one-way quantum computer – a non-network model of quantum computation,” *Journal of Modern Optics*, vol. 49, no. 8, pp. 1299–1306, Jul. 2002. DOI: [10.1080/09500340110107487](https://doi.org/10.1080/09500340110107487). arXiv: [quant-ph/0108118](https://arxiv.org/abs/quant-ph/0108118).
- [39] R. Raussendorf and H. Briegel. “Computational model underlying the one-way quantum computer.” arXiv: [quant-ph/0108067](https://arxiv.org/abs/quant-ph/0108067). (Mar. 12, 2002), [Online]. Available: <http://arxiv.org/abs/quant-ph/0108067>, pre-published.
- [40] R. Raussendorf, D. E. Browne, and H. J. Briegel, “Measurement-based quantum computation with cluster states,” *Physical Review A*, vol. 68, no. 2, p. 022312, Aug. 25, 2003. DOI: [10.1103/PhysRevA.68.022312](https://doi.org/10.1103/PhysRevA.68.022312). arXiv: [quant-ph/0301052](https://arxiv.org/abs/quant-ph/0301052).
- [41] M. A. Nielsen, “Cluster-state quantum computation,” *Reports on Mathematical Physics*, vol. 57, no. 1, pp. 147–161, Feb. 2006. DOI: [10.1016/S0034-4877\(06\)80014-5](https://doi.org/10.1016/S0034-4877(06)80014-5). arXiv: [quant-ph/0504097](https://arxiv.org/abs/quant-ph/0504097).
- [42] D. Greenberger, *Compendium of Quantum Physics: Concepts, Experiments, History and Philosophy*. Berlin, Heidelberg: Springer Berlin / Heidelberg, 2009, 1 p.
- [43] D. E. Browne and T. Rudolph, “Resource-efficient linear optical quantum computation,” *Physical Review Letters*, vol. 95, no. 1, p. 010501, Jun. 27, 2005. DOI: [10.1103/PhysRevLett.95.010501](https://doi.org/10.1103/PhysRevLett.95.010501). arXiv: [quant-ph/0405157](https://arxiv.org/abs/quant-ph/0405157).
- [44] F. Ewert and P. van Loock. “3/4-efficient Bell measurement with passive linear optics and unentangled ancillae,” arXiv.org. (Mar. 19, 2014), [Online]. Available: <https://arxiv.org/abs/1403.4841v1>.
- [45] M. Zwerger, A. Pirker, V. Dunjko, H. J. Briegel, and W. Dür, “Long-range big quantum-data transmission,” *Physical Review Letters*, vol. 120, no. 3, p. 030503, Jan. 19, 2018. DOI: [10.1103/PhysRevLett.120.030503](https://doi.org/10.1103/PhysRevLett.120.030503). arXiv: [1705.02174](https://arxiv.org/abs/1705.02174) [quant-ph].
- [46] W. Dür, H.-J. Briegel, J. I. Cirac, and P. Zoller, “Quantum repeaters based on entanglement purification,” *Physical Review A*, vol. 59, no. 1, pp. 169–181, Jan. 1, 1999. DOI: [10.1103/PhysRevA.59.169](https://doi.org/10.1103/PhysRevA.59.169). arXiv: [quant-ph/9808065](https://arxiv.org/abs/quant-ph/9808065).
- [47] S. Krastanov, V. V. Albert, and L. Jiang, “Optimized Entanglement Purification,” *Quantum*, vol. 3, p. 123, Feb. 18, 2019. DOI: [10.22331/q-2019-02-18-123](https://doi.org/10.22331/q-2019-02-18-123). arXiv: [1712.09762](https://arxiv.org/abs/1712.09762) [quant-ph].
- [48] W. Dür, H. Aschauer, and H.-J. Briegel, “Multiparticle entanglement purification for graph states,” *Physical Review Letters*, vol. 91, no. 10, p. 107903, Sep. 5, 2003. DOI: [10.1103/PhysRevLett.91.107903](https://doi.org/10.1103/PhysRevLett.91.107903). arXiv: [quant-ph/0303087](https://arxiv.org/abs/quant-ph/0303087).
- [49] C. Kruszynska, A. Miyake, H. J. Briegel, and W. Dür, “Entanglement purification protocols for all graph states,” *Physical Review A*, vol. 74, no. 5, p. 052316, Nov. 10, 2006. DOI: [10.1103/PhysRevA.74.052316](https://doi.org/10.1103/PhysRevA.74.052316). arXiv: [quant-ph/0606090](https://arxiv.org/abs/quant-ph/0606090).
- [50] M. Varnava, D. E. Browne, and T. Rudolph, “How good must single photon sources and detectors be for efficient linear optical quantum computation?” *Physical Review Letters*, vol. 100, no. 6, p. 060502, Feb. 12, 2008. DOI: [10.1103/PhysRevLett.100.060502](https://doi.org/10.1103/PhysRevLett.100.060502). arXiv: [quant-ph/0702044](https://arxiv.org/abs/quant-ph/0702044).
- [51] J. Borregaard, H. Pichler, T. Schröder, M. D. Lukin, P. Lodahl, and A. S. Sørensen, “One-Way Quantum Repeater Based on Near-Deterministic Photon-Emitter Interfaces,” *Physical Review X*, vol. 10, no. 2, p. 021071, Jun. 30, 2020. DOI: [10.1103/PhysRevX.10.021071](https://doi.org/10.1103/PhysRevX.10.021071).
- [52] B. Li, S. E. Economou, and E. Barnes, “Photonic resource state generation from a minimal number of quantum emitters,” *npj Quantum Information*, vol. 8, no. 1, p. 11, Feb. 3, 2022. DOI: [10.1038/s41534-022-00522-6](https://doi.org/10.1038/s41534-022-00522-6). arXiv: [2108.12466](https://arxiv.org/abs/2108.12466) [quant-ph].
- [53] S. Ghanbari, J. Lin, B. MacLellan, L. Robichaud, P. Roztock, and H.-K. Lo, “Optimization of deterministic photonic-graph-state generation via local operations,” *Physical Review A*, vol. 110, no. 5, p. 052605, Nov. 15, 2024. DOI: [10.1103/PhysRevA.110.052605](https://doi.org/10.1103/PhysRevA.110.052605).
- [54] P. Thomas, L. Ruscio, O. Morin, and G. Rempe, “Fusion of deterministically generated photonic graph states,” *Nature*, vol. 629, no. 8012, pp. 567–572, May 2024. DOI: [10.1038/s41586-024-07357-5](https://doi.org/10.1038/s41586-024-07357-5).
- [55] A. Patil and S. Guha. “An Improved Design for All-Photonic Quantum Repeaters.” arXiv: [2405.11768](https://arxiv.org/abs/2405.11768) [quant-ph]. (May 20, 2024), [Online]. Available: <http://arxiv.org/abs/2405.11768>, pre-published.
- [56] B. Li, K. Goodenough, F. Rozpedek, and L. Jiang, “Generalized Quantum Repeater Graph States,” *Physical Review Letters*, vol. 134, no. 19, p. 190801, May 12, 2025. DOI: [10.1103/PhysRevLett.134.190801](https://doi.org/10.1103/PhysRevLett.134.190801). arXiv: [2407.01429](https://arxiv.org/abs/2407.01429) [quant-ph].

- [57] Y. Zhan, P. Hilaire, E. Barnes, S. E. Economou, and S. Sun, “Performance analysis of quantum repeaters enabled by deterministically generated photonic graph states,” *Quantum*, vol. 7, p. 924, Feb. 16, 2023. DOI: [10.22331/q-2023-02-16-924](https://doi.org/10.22331/q-2023-02-16-924).
- [58] N. Benchasattabuse. “Resource Management in Heterogeneous Quantum Repeater Networks.” version 1. (2026), [Online]. Available: <https://arxiv.org/abs/2605.25132>, pre-published.
- [59] N. Benchasattabuse, M. Hajdušek, and R. Van Meter, “Architecture and Protocols for All-Photonic Quantum Repeater,” in *2024 IEEE International Conference on Quantum Computing and Engineering (QCE)*, Montreal, QC, Canada: IEEE, Sep. 15, 2024, pp. 1879–1889. DOI: [10.1109/QCE60285.2024.00217](https://doi.org/10.1109/QCE60285.2024.00217).
- [60] W. J. Munro, K. Azuma, K. Tamaki, and K. Nemoto, “Inside Quantum Repeater,” *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 21, no. 3, pp. 78–90, May 2015. DOI: [10.1109/JSTQE.2015.2392076](https://doi.org/10.1109/JSTQE.2015.2392076).
- [61] G. X. Yau, A. Burushkina, F. F. da Silva, S. Maji, P. S. Thomas, and G. Vardoyan. “Reinforcement Learning for Quantum Network Control with Application-Driven Objectives.” arXiv: [2509.10634](https://arxiv.org/abs/2509.10634) [quant-ph]. (Mar. 30, 2026), [Online]. Available: <http://arxiv.org/abs/2509.10634>, pre-published.
- [62] R. V. Meter, K. Nemoto, and W. J. Munro, “Communication Links for Distributed Quantum Computation,” *IEEE Transactions on Computers*, vol. 56, no. 12, pp. 1643–1653, Dec. 2007. DOI: [10.1109/TC.2007.70775](https://doi.org/10.1109/TC.2007.70775). arXiv: [quant-ph/0701043](https://arxiv.org/abs/quant-ph/0701043).
- [63] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary edition. Cambridge: Cambridge university press, 2010.
- [64] R. Sethi and J. D. Ullman, “The Generation of Optimal Code for Arithmetic Expressions,” *Journal of the ACM (JACM)*, vol. 17, no. 4, pp. 715–728, Oct. 1, 1970. DOI: [10.1145/321607.321620](https://doi.org/10.1145/321607.321620).
- [65] M. L. Puterman, *Markov Decision Processes: Discrete Stochastic Dynamic Programming* (Wiley Series in Probability and Statistics v.414). Hoboken: John Wiley & Sons, Inc, 2009, 680 pp.
- [66] R. S. Sutton and A. Barto, *Reinforcement Learning: An Introduction* (Adaptive Computation and Machine Learning), Nachdruck. Cambridge, Massachusetts: The MIT Press, 2014, 322 pp.
- [67] A. Jamiolkowski, “Linear transformations which preserve trace and positive semidefiniteness of operators,” *Reports on Mathematical Physics*, vol. 3, no. 4, pp. 275–278, Dec. 1, 1972. DOI: [10.1016/0034-4877\(72\)90011-0](https://doi.org/10.1016/0034-4877(72)90011-0).
- [68] M.-D. Choi, “Completely positive linear maps on complex matrices,” *Linear Algebra and its Applications*, vol. 10, no. 3, pp. 285–290, Jun. 1, 1975. DOI: [10.1016/0024-3795\(75\)90075-0](https://doi.org/10.1016/0024-3795(75)90075-0).
- [69] V. Pareto, *Manual of Political Economy: A Critical and Variorum Edition*, A. Montesano, A. Zanni, L. Bruni, and J. S. Chipman, Eds. Oxford, United Kingdom: Oxford Univ Pr on Demand, 2014, 664 pp.
- [70] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, “A fast and elitist multiobjective genetic algorithm: NSGA-II,” *IEEE Transactions on Evolutionary Computation*, vol. 6, no. 2, pp. 182–197, Apr. 2002. DOI: [10.1109/4235.996017](https://doi.org/10.1109/4235.996017).
- [71] M. Ehrgott, *Multicriteria Optimization*. Springer Science & Business Media, May 18, 2005, 329 pp. Google Books: [8wGyB5Sa2CUC](https://books.google.com/books?id=8wGyB5Sa2CUC).
- [72] R. Bellman, *Dynamic Programming*. Mineola, N.Y: Dover Publications, 2003, 384 pp.
- [73] B. Lowerre and R. Reddy, “THE HARP SPEECH UNDERSTANDING SYSTEM,” in *Readings in Speech Recognition*, Elsevier, 1990, pp. 576–586. DOI: [10.1016/B978-0-08-051584-7.50053-X](https://doi.org/10.1016/B978-0-08-051584-7.50053-X).
- [74] L. Huang, K. Zhao, and M. Ma. “When to Finish? Optimal Beam Search for Neural Text Generation (modulo beam size).” arXiv: [1809.00069](https://arxiv.org/abs/1809.00069) [cs.CL]. (Aug. 31, 2018), [Online]. Available: <http://arxiv.org/abs/1809.00069>, pre-published.
- [75] C. Meister, T. Vieira, and R. Cotterell. “Best-First Beam Search.” arXiv: [2007.03909](https://arxiv.org/abs/2007.03909) [cs.CL]. (Nov. 15, 2022), [Online]. Available: <http://arxiv.org/abs/2007.03909>, pre-published.
- [76] E. Flechtner, *Entanglement Purification Scheduler*, version v1, Sep. 15, 2026. [Online]. Available: <https://github.com/EliottFlechtner/entanglement-purification-scheduler>.

Appendix A

Additional Background and Technical Details

A.1 Repeater Operations and Purification

This section collects illustrative figures supporting Chapter 3: the elementary Bell-state measurement, a worked example of backbone selection across a full chain, and the circuit-level implementation of the three purification circuits.

A.1.1 Bell-State Measurement

Section 3.1.1 introduces the Bell-state measurement (BSM) as the elementary operation used to establish and extend entanglement links. Figure A.1 illustrates it in more detail than the main text.

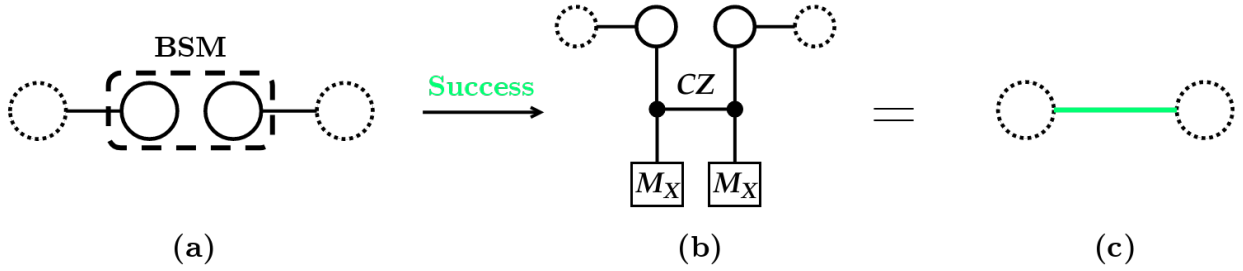


Figure A.1: Entanglement swapping via a linear-optical Bell-State Measurement (BSM). (a) Two incoming photonic qubits from neighbouring resources interfere at a beamsplitter and are measured in the computational basis. Only two of the four Bell states are distinguishable, limiting the success probability to 50% per attempt (Section 3.1.1). (b) Graph-state representation of a successful BSM, equivalent to a CZ gate followed by X -basis measurements (M_X) on the measured qubits. (c) The resulting edge (green) directly connects the two remaining qubits, establishing entanglement between the neighbouring resources.

A.1.2 Backbone Selection Example

Section 3.1.3 describes *backbone selection*: at every ABSA, the first arm whose BSM succeeds is kept and every other arm at that hop is discarded, regardless of which arm that turns out to be. Figure A.2 works through a concrete 3-hop chain to make this concrete: each ABSA independently keeps whichever of its arms first succeeds, and the surviving entanglement (bottom row) connects Alice to Bob only through the chain of kept arms, one hop at a time, regardless of which specific arm succeeded at each hop.

A.1.3 Purification Circuits

Section 3.3.1 introduces the three purification circuits $\mathcal{C}_{ZX}$, $\mathcal{C}_{XZ}$, and $\mathcal{C}_{YY}$ by the stabilizer each one measures. Figure A.3 below gives their circuit-level implementation.

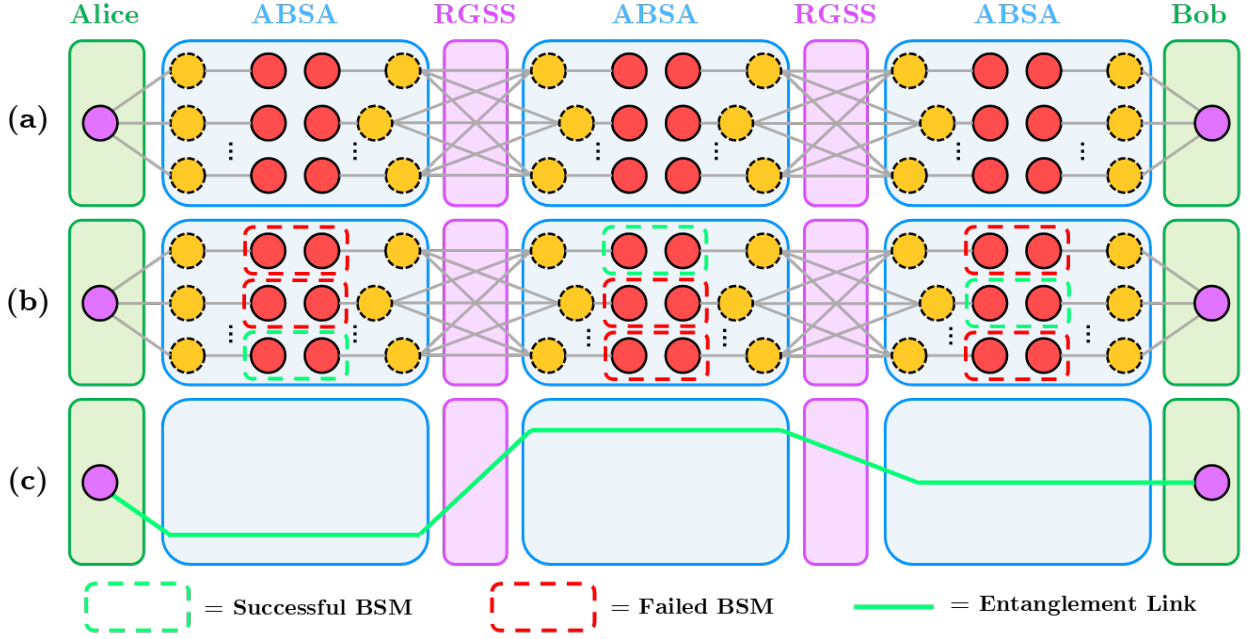


Figure A.2: Backbone selection across a $N = 3$ -hop chain. **(a)** The network configuration before entanglement generation. Each ABSA initially holds outer/inner qubit pairs (red/yellow) for all provisioned arms connecting it to each of its two neighbouring stations. **(b)** At each ABSA, a BSM is attempted for every available arm. Multiple attempts may succeed, but once at least one successful BSM is obtained (green), one successful arm is selected and its corresponding inner qubits are retained, while the remaining arms are discarded. As this selection is performed independently at each ABSA, the retained arms need not correspond to the same arm index across the different hops. **(c)** The selected inner qubits form the backbone used to establish end-to-end entanglement. The resulting entanglement (green line) connects Alice to Bob through one retained arm at each hop, as described in Section 3.1.3.

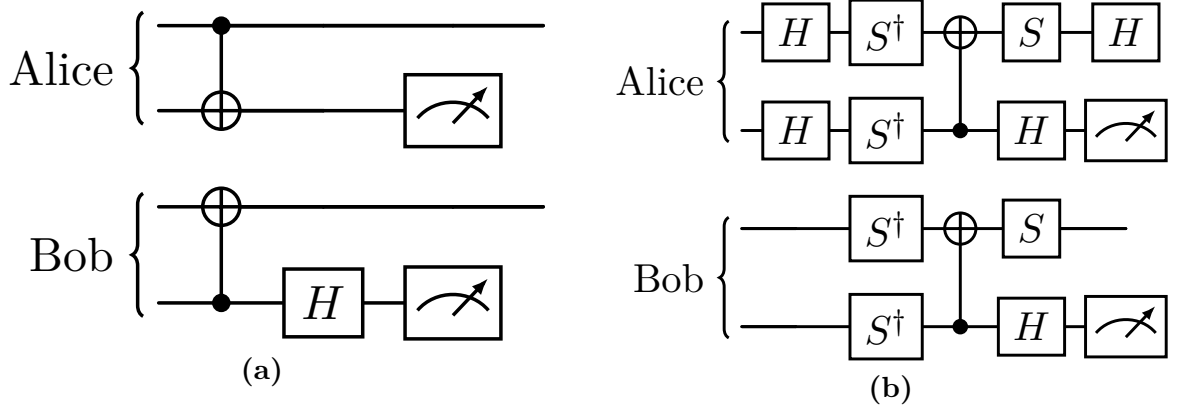


Figure A.3: Circuit-level implementation of two of the purification circuits introduced in Section 3.3.1. Each circuit consumes two same-span resources, one held by Alice and one by Bob with top (bottom) wire per party is the retained (measured) qubit respectively. It produces one purified resource on the retained wires when the two parties' measurement outcomes agree. **(a)** C_{ZX} : Alice measures the Z -parity of her pair while Bob measures the X -parity of his. The success probability and output error vector is given previously in Eqs. (3.4) and (3.7). The companion circuit C_{XZ} (not shown) is identical to C_{ZX} but with Alice's and Bob's local circuits swapped, so that Alice measures the X -parity and Bob the Z -parity (Eqs. (3.5), (3.8)). **(b)** C_{YC} : Alice applies a local $H-S^\dagger$ basis change on both her qubits before her CNOT, while Bob applies only a local $S^\dagger$ basis change (without H) on both his qubits before his CNOT, so that the compared parity is a joint Y -type stabilizer across all four qubits (see Eqs. (3.6), (3.9)).

A.2 Pseudocode and Concurrency Analysis

Section 4.4 defines the concurrency requirement $M(\Sigma)$ and its Sethi–Ullman recurrence [64]. This appendix gives the pseudocode implementing that recurrence as well as a complexity analysis.

The following pseudocode implements this recurrence. It assumes that the schedule is a binary tree and that each node exposes its children through $\text{children}(v)$.

Algorithm 1: Minimum concurrency requirement of a tree-shaped schedule

```

1 Function RegisterRequirement( $v$ ):
2   if  $v$  is GenNode then
3     return 1
4   if  $v$  is IdleNode or  $v$  is HeraldNode then
5      $c \leftarrow \text{child}(v)$ 
6     return RegisterRequirement( $c$ )
7    $(c_1, c_2) \leftarrow \text{children}(v)$ 
8    $L \leftarrow \text{RegisterRequirement}(c_1)$ 
9    $R \leftarrow \text{RegisterRequirement}(c_2)$ 
10  if  $L \neq R$  then
11    return  $\max(L, R)$ 
12  else
13    return  $L + 1$ 

```

The procedure is evaluated recursively from the root, so every node is visited exactly once. Consequently, the computation takes $O(|T|)$ time and $O(h)$ auxiliary space for a tree of $|T|$ nodes and height h . The resulting value is thus the concurrency requirement $M(\Sigma) = \text{REGISTERREQUIREMENT}(\text{root}(T))$. The schedule satisfies the concurrency budget when $M(\Sigma) \leq M_{\max}$.

Appendix B

Additional Experimental Results

This appendix collects material supporting Chapter 6: extended numeric tables and figures behind results shown in the main text, together with supplementary diagnostics for the search algorithms of Chapter 5.

B.1 Experimental Parameters

Table B.1 lists every parameter varied in Chapter 6, together with the reference value used when it is held fixed and the section in which it is varied.

Parameter	Symbol	Reference ($\mathcal{N}_{\text{ref}}$)	Swept ranges	Where varied
Hop count	N	10	10–28	Section 6.5.1
Hop length	ℓ_i	2 km	{1, 2, 3, 4} km	Section 6.3.3
Tree-encoding branching vector	$\vec{b}^{(i)}$	(16, 14, 1)	–	–
BSM arm count	k_i	18	6, 18	Section 6.3.2
Inner-qubit error rates	$\{p_X^{\text{in}}, p_Z^{\text{in}}\}_i$	0	$[0, 3 \times 10^{-3}]$	Section 6.3.2
Photon survival probability	η_i	0.912	$10^{-0.2\ell_i/10}$	Section 6.3.3
Outer-photon depolarizing rate	e_d	0.01	$[0, 0.01]$	Section 6.3.4
Memory dephasing rate	γ	0	$[0, 10^5]$	Section 6.6
Propagation speed	c	2×10^5 km/s	–	–
Emission delay	τ_{emit}	0	$[10^{-4}, 10^0]$	Section 6.6
Generation-cost budget	E_{max}	100	50–280	Sections 6.3.1–6.5.1

Table B.1: Experimental parameters. The reference column gives the nominal value used unless a parameter is varied in the indicated experiment. All parameters not listed as varying follow the Integrating paper’s [17] reference configuration $\mathcal{N}_{\text{ref}}$ (Section 4.1). The tree branching vector is not varied because it does not affect the objectives F , C , or R introduced in Chapter 4. Uniform hop-length scaling likewise leaves relative schedule comparisons unchanged; heterogeneous hop lengths are considered separately in Section 6.3.3.

B.2 Extended Physics Validation

Section 6.2 reports the evaluator’s agreement with the fidelity results of the Integrating paper’s Fig. 5 [17] when varying the decoherence parameter e_d . Table B.2 and Figure B.1 provide the evaluator’s fidelity output for the raw, heralded-pumping, and optimistic-pumping canonical schedules (Section 3.2.3) across the full range $e_d \in [0, 0.01]$ used throughout Chapter 6.

The source paper’s Fig. 5 also covers this full e_d range, and Figure B.1 reproduces its shape throughout. The paper reports numeric fidelity values at only two points on the curve, $e_d = 0.000$ and $e_d = 0.010$: approximately {1.000, 1.000, 1.000} and {0.823, 0.917, 0.929}, respectively, for the same three schedules in the same order. The evaluator matches both published points to at least three significant figures, and to four at $e_d = 0.010$ (already reported in Section 6.2 and shown in bold in Table B.2). The remaining nine rows trace the same reproduced curve and are included for completeness.

e_d	F_{raw}	$F_{\text{baseline (heralded)}}$	$F_{\text{flexible (optimistic)}}$
0.000	1.0000	1.0000	1.0000
0.001	0.9803	0.9932	0.9933
0.002	0.9610	0.9861	0.9866
0.003	0.9422	0.9787	0.9799
0.004	0.9239	0.9710	0.9730
0.005	0.9061	0.9629	0.9661
0.006	0.8887	0.9544	0.9591
0.007	0.8717	0.9456	0.9519
0.008	0.8552	0.9364	0.9446
0.009	0.8391	0.9268	0.9372
0.010	0.8234	0.9168	0.9295

Table B.2: Evaluator-derived fidelity vs. e_d for the three canonical schedules of Section 3.2.3, at $N = 10$. Bold rows ($e_d = 0.000$ and $e_d = 0.010$) are the only two points with a published reference value to compare against (see text).

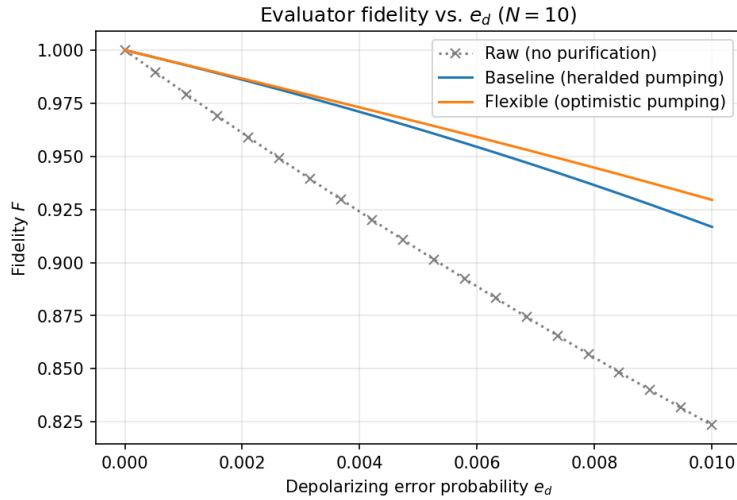


Figure B.1: Evaluator fidelity vs. e_d for the three canonical schedules of Table B.2, $N = 10$, reproducing the shape of the source paper’s Fig. 5 [17]. The raw chain degrades fastest. Both pumping variants remain close, with the optimistic variant consistently slightly above the heralded variant over the full range (Section 7.3).

B.3 Extended Network and Heterogeneity Results

This section provides the full numeric results for the two network-generalization experiments of Section 6.3: uniformly applied non-ideal parameters (Section 6.3.2) and per-hop heterogeneous networks (Section 6.3.3).

B.3.1 Network Generalization Comparison

The network-generalization experiment of Section 6.3.2 is reported in full in Table B.3, which gives the cost, fidelity, and rate values underlying Figure 6.1 in the main text for both tested BSM arm counts.

B.3.2 Heterogeneous Weak-Link Contrast

Section 6.3.3 considers both a deliberately constructed five-hop weak-link network and a sample of 100 randomly generated networks. Table B.4 gives the complete numeric comparison for the designed case, in which one hop has an inner-qubit error rate 12 times higher than those of its neighbours.

Figure B.2 shows the complete 100-network random sample. Each point gives the optimizer’s rate

Configuration	Schedule	C	F	Rate	Feasible
18 arms	Paper schedule	100	0.67	103.18	no
18 arms	Matched-cost	100	0.90	25.31	yes
18 arms	Budget-relaxed	80	0.90	85.30	yes
6 arms	Paper schedule	100	0.87	871.55	no
6 arms	Matched-cost	100	0.93	681.56	yes
6 arms	Budget-relaxed	60	0.91	2096.91	yes

Table B.3: Optimizer versus the paper schedule under two non-ideal network configurations. Both use $N = 10$, $e_d = 0.01$, $E_{\max} = 100$, $\gamma = 0$, and non-zero inner-qubit error $p_X^{\text{in}} = p_Z^{\text{in}} = 3 \times 10^{-3}$, unlike the source paper’s idealized inner-qubit assumption. The configurations differ in the number of redundant BSM arms per hop. A schedule is feasible when $F \geq F_{\min} = 0.9$.

Schedule	C	F	Rate	Feasible
Best uniform link-level recipe	40	0.88	1738.32	no
Optimizer’s schedule	38	0.90	1798.57	yes

Table B.4: Designed weak-link contrast for a heterogeneous $N = 5$ network at $e_d = 0.01$, $E_{\max} = 50$, and $F_{\min} = 0.9$. One hop has an inner-qubit error rate $12\times$ higher than its neighbours. The uniform link-level baseline uses the same purification configuration at every hop, whereas the optimizer may allocate resources differently across hops. A schedule is feasible when $F \geq F_{\min}$.

improvement over the best feasible uniform link-level recipe as a function of the network’s per-hop noise heterogeneity, measured by the coefficient of variation (i.e. the standard deviation divided by the mean inner-qubit error rate across hops). The largest improvements occur in networks with relatively modest noise dispersion, where a single noisier hop leaves the remaining hops sufficiently inexpensive for the optimizer to reallocate resources towards the weak link.

B.4 Extended Objective and Optimality Results

B.4.1 Alternative Objectives Comparison

The objective-substitution experiment of Section 6.4.1 is detailed in Table B.5, which reports the full five-objective comparison at $N = 10$ and $e_d = 0.01$.

Objective	C	F	Rate
Maximize rate	60	0.9063	6195.95
Maximize fidelity, paper-rate floor	80	0.9351	4804.58
Maximize fidelity, matched-rate floor	80	0.9351	4804.58
Minimize cost, $F \geq 0.9$ only	60	—	1239.19
Minimize cost, $F \geq 0.9$ and $R \geq 4804.58$	60	—	6195.95

Table B.5: Objective-substitution comparison at $N = 10$, $e_d = 0.01$: five objective definitions with no change to the search algorithm.

B.5 Search Algorithm Benchmarks

This section first provides additional evidence for the comparable scaling of the exact dynamic program once same-span pumping is enabled (Section 5.4). Second, it presents runtime benchmarks for representative chain lengths, highlighting cases where the pumping-enabled exact dynamic program does not provide a strict upper bound on beam-search runtime.

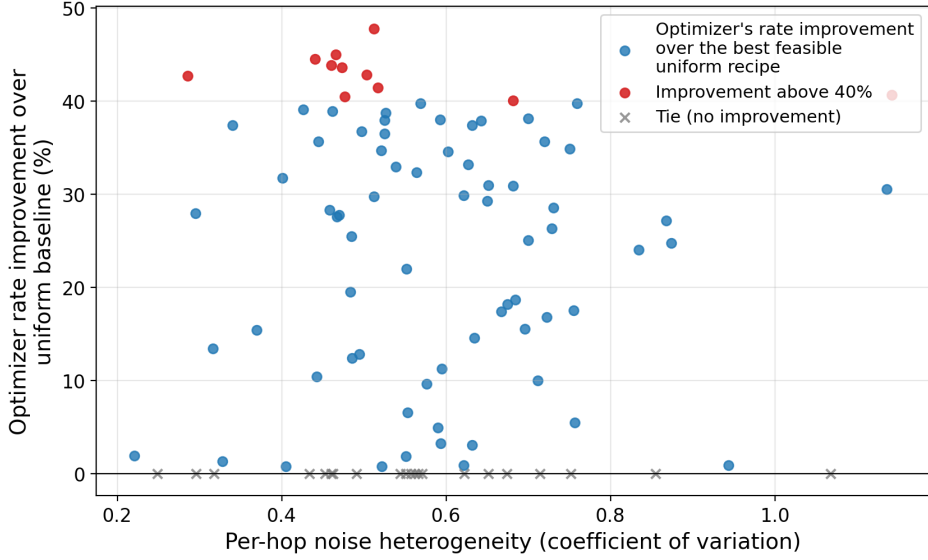


Figure B.2: Optimizer rate improvement over the best feasible uniform link-level recipe vs. per-hop noise heterogeneity, across the 100 random $N = 6$ networks of Section 6.3.3. Grey crosses on the zero line are exact ties, where the optimizer’s own best schedule coincides with the uniform recipe; these represent no advantage, not a loss (Section 7.1). Red points mark the networks where the improvement exceeds 40%, the upper end of the observed range.

B.5.1 Runtime Benchmarks

Table B.6 reports the wall-clock time and returned score underlying Figure B.3 and the beam-width analysis for representative chain lengths. It also identifies the two cases, footnoted in the table, where the pumping-enabled exact dynamic program does not provide a strict upper bound on beam-search runtime (Section 5.4).

N	$E_{\max}$	Exact DP (s)	Beam search (s)	Exact DP score	Beam search score
2	20	0.10	0.10	50 000	50 000
6	60	3.82	3.34	14 372	15 391 [‡]
10	100	20.43	16.50	6 196	6 196
14	140	85.37	66.93	3 394	3 394
18	180	203.68	153.52	$-\infty$ [†]	1 214

Table B.6: Wall-clock runtime and returned score for the exact dynamic program and beam search at representative chain lengths. Runs use $E_{\max} = 10N$, $F_{\min} = 0.9$, beam width $w_{\text{beam}} = 25$, same-span pumping enabled (Section 5.4), a 300 s timeout, and a 3 GiB memory limit. The score is the value of the active objective; here, it is the returned schedule rate $R(\Sigma)$. [†]At $N = 18$, the exact dynamic program completes within the timeout but finds no feasible candidate, while beam search returns a feasible one. [‡]At $N = 6$, beam search returns a higher score than the exact dynamic program. These cases arise because same-span pumping is itself beam-limited, so the pumping-enabled exact dynamic program is not guaranteed to upper-bound beam search (Section 5.4).

B.5.2 Runtime vs. Chain Length

Figure B.3 extends the runtime comparison of Table B.6 (Section B.5.1) to every tested N . Once the same-span pumping move is enabled, the exact dynamic program and beam search exhibit similar scaling with N , while beam search remains consistently faster.

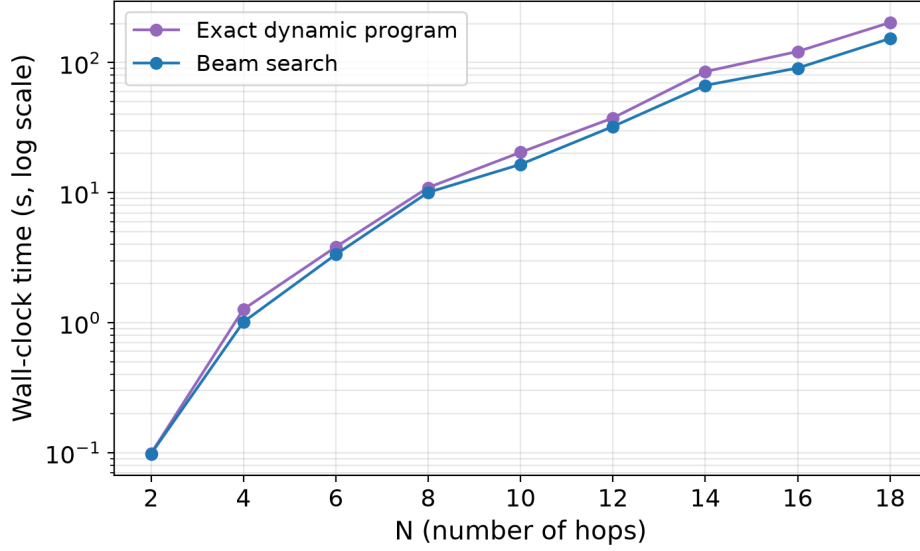


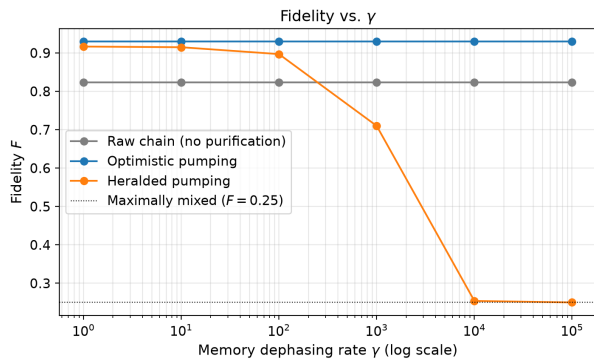
Figure B.3: Search wall-clock time vs. N for the exact dynamic program and beam search, with the same-span pumping move enabled throughout.

B.6 Timing Sensitivity Sweeps

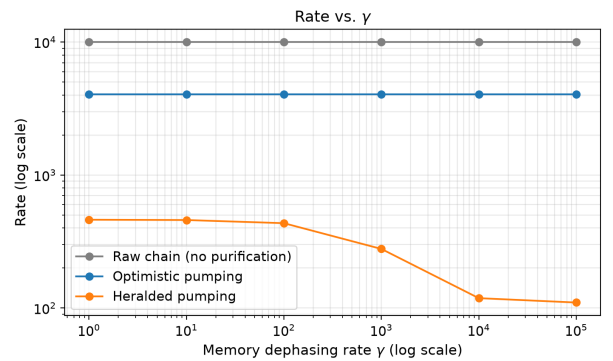
Section 6.6 examines the effects of memory dephasing and generation latency on schedule rate and fidelity. Figure B.4 shows both sweeps in full at $N = 10$ and $e_d = 0.01$.

Memory dephasing (γ) Increasing γ from 0 to 10^5 leaves the raw chain and paper schedule unchanged, as neither requires resources to wait for another branch. In contrast, heralded end-node pumping drops from $F = 0.92$ to $F = 0.25$ as its sacrificial copies accumulate dephasing during repeated heralding delays. The optimizer’s schedules remain stable across the sweep, favouring optimistic schedules that avoid these storage intervals (Section 7.3).

Generation latency (τ_{emit}) Increasing τ_{emit} at $\gamma = 0$ reduces the rate for all schedules. The relative loss is largest for optimistic pumping, with a baseline latency of about $1 \times L/c$, and smallest for heralded pumping, with about $9 \times L/c$. The two cross near $\tau_{\text{emit}} \approx 0.01$, above which the larger fixed latency of heralded pumping makes it less sensitive to further increases.



(a) Fidelity vs. γ



(b) Rate vs. γ (log-log)

Figure B.4: Timing sensitivity of the raw chain, the paper schedule, and heralded end-node pumping at $N = 10$ and $e_d = 0.01$. **(a)** Fidelity as a function of the memory dephasing rate γ for the three schedules, showing the strong degradation of heralded pumping at large γ . **(b)** Corresponding rate as a function of γ on a log-log scale; the raw chain and optimistic pumping remain constant, while the rate of heralded pumping decreases as memory dephasing increases.